

RANDOM VARIABLES

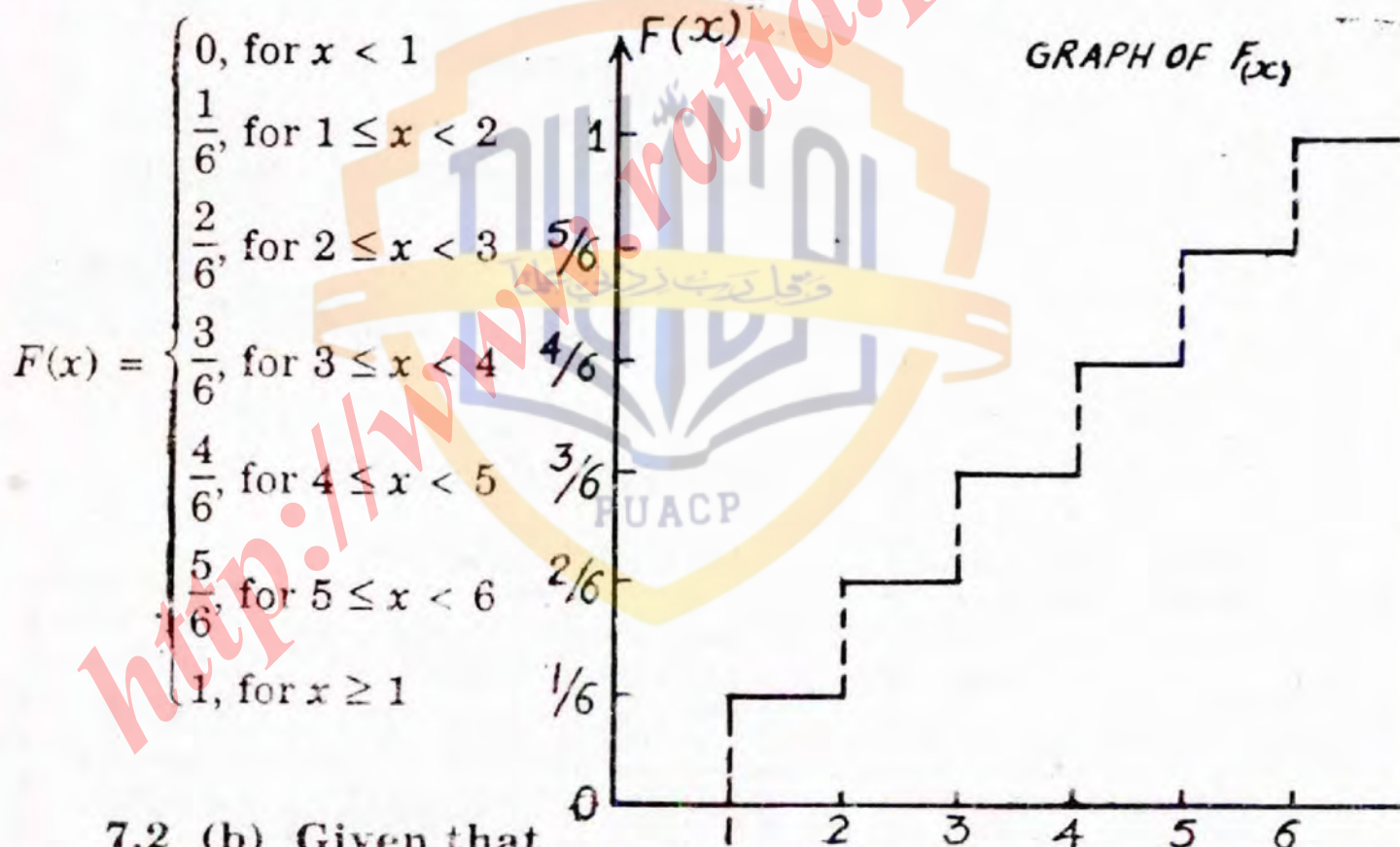
7.1. (b) The sample space for this experiment is

$$S = \{1, 2, 3, 4, 5, 6\}$$

Let X be the r.v. that denote the number of points appearing. Then the values of x are 1, 2, 3, 4, 5 and 6. As all six faces are equally likely to appear, therefore

$$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = 1/6$$

Hence the desired distribution function, $F(x)$, is



7.2 (b) Given that

x	-1	0	1	Total
$f(x)$	$3c$	$3c$	$6c$	$12c$

(i) Since the sum of probabilities in a *p.d.* should be equal to 1, therefore $12c = 1$ or $c = \frac{1}{12}$.

(ii) The probability distribution of $Y = 2X + 1$ is found as below:

x_i	-1	0	1
$f(x_i)$	$3c = \frac{3}{12}$	$\frac{3}{12}$	$\frac{6}{12}$
$y_i = 2x_i + 1$	-1	1	3

Hence the desired p.d. of $Y = 2X + 1$ is

y_i	-1	1	3
$f(y_i)$	$3/12$	$3/12$	$6/12$

(c) Let X be the random variable that denotes the number of aces in a bridge hand. Then the values that X can take, are 0, 1, 2, 3, and 4. The desired p.d. is given below:

x_i	$f(x_i)$
0	$\frac{\binom{4}{0} \binom{48}{13}}{\binom{52}{13}} = \frac{6327}{20825} = 0.3038$
1	$\frac{\binom{4}{1} \binom{48}{12}}{\binom{52}{13}} = \frac{9139}{20825} = 0.4389$
2	$\frac{\binom{4}{2} \binom{48}{11}}{\binom{52}{13}} = \frac{4446}{20825} = 0.2135$
3	$\frac{\binom{4}{3} \binom{48}{10}}{\binom{52}{13}} = \frac{858}{20825} = 0.0412$
4	$\frac{\binom{4}{4} \binom{48}{9}}{\binom{52}{13}} = \frac{55}{20825} = 0.0026$
Total	1

7.3. (b) S contains $\binom{10}{4} = 210$ sample points.

Let X be the r.v. that denotes the number of red balls. Then the values of X are 0, 1, 2, 3 and 4, and their probabilities are

$$f(0) = P(X=0) = \frac{\binom{4}{0} \binom{6}{4}}{\binom{10}{4}} = \frac{15}{210}$$

$$f(1) = P(X=1) = \frac{\binom{4}{1} \binom{6}{3}}{\binom{10}{4}} = \frac{80}{210}$$

$$f(2) = P(X=2) = \frac{\binom{4}{2} \binom{6}{2}}{\binom{10}{4}} = \frac{90}{210}$$

$$f(3) = P(X=3) = \frac{\binom{4}{3} \binom{6}{1}}{\binom{10}{4}} = \frac{24}{210}$$

$$f(4) = P(X=4) = \frac{\binom{4}{4} \binom{6}{0}}{\binom{10}{4}} = \frac{1}{210}$$

Putting this information in a tabular form, we obtain the desired probability distribution of X as

No. of red balls: x_i	0	1	2	3	4
Probability: $f(x_i)$	$\frac{15}{210}$	$\frac{80}{210}$	$\frac{90}{210}$	$\frac{24}{210}$	$\frac{1}{210}$

7.4. (a) S consists of $\binom{15}{3} = 455$ sample points.

Let X be the r.v. that denotes the number of defective clock radios in the sample. Then the values of x are 0, 1, 2, 3 and their probabilities are

$$f(0) = P(X=0) = \frac{\binom{5}{0} \binom{10}{3}}{\binom{15}{3}} = \frac{120}{455}$$

$$f(1) = P(X=1) = \frac{\binom{5}{1} \binom{10}{2}}{\binom{15}{3}} = \frac{225}{455}$$

$$f(2) = P(X=2) = \frac{\binom{5}{2} \binom{10}{1}}{\binom{15}{3}} = \frac{100}{455}$$

$$f(3) = P(X=3) = \frac{\binom{5}{3} \binom{10}{0}}{\binom{15}{3}} = \frac{10}{455}$$

Hence the desired probability distribution of X is

No. of defectives: x_i	0	1	2	3	Total
Probability: $f(x_i)$	$\frac{120}{455}$	$\frac{225}{455}$	$\frac{100}{455}$	$\frac{10}{455}$	1

(b) S consists of $\binom{8}{3} = 56$ sample points.

Let X be the r.v. that denotes the number of white balls, drawn from the bag. Then the values of x are 0, 1, 2, 3 and their probabilities are:

$$f(0) = P(X=0) = \frac{\binom{5}{0} \binom{3}{3}}{\binom{8}{3}} = \frac{1}{56}$$

$$f(1) = P(X=1) = \frac{\binom{5}{1} \binom{3}{2}}{\binom{8}{3}} = \frac{15}{56}$$

$$f(2) = P(X=2) = \frac{\binom{5}{2} \binom{3}{1}}{\binom{8}{3}} = \frac{30}{56}$$

$$f(3) = P(X=3) = \frac{\binom{5}{3} \binom{3}{0}}{\binom{8}{3}} = \frac{10}{56}$$

Putting this information in a tabular form, we obtain the desired probability distribution of X as

No. of white balls: x_i	0	1	2	3	Total
Probability: $f(x_i)$	$\frac{1}{56}$	$\frac{15}{56}$	$\frac{30}{56}$	$\frac{10}{56}$	1

7.5. (b) The function $f(x)$ will be a density function, if

$$(i) \quad f(x) \geq 0, \text{ and } (ii) \quad \int_{-\infty}^{\infty} f(x) dx = 1$$

The first condition is satisfied for every x in the given range, and the second condition will be satisfied if

$$A \int_0^2 (4x - 2x^2) dx = 1.$$

$$\text{i.e. if } A \left\{ \left[\frac{4x^2}{2} \right]_0^2 - 2 \left[\frac{x^3}{3} \right]_0^2 \right\} = 1,$$

$$\text{i.e. if } A \left\{ 8 - 2 \left(\frac{8}{3} \right) \right\} = 1,$$

$$\text{i.e. if } A \left\{ \frac{24 - 16}{3} \right\} = 1.$$

This gives $A = \frac{3}{8}$.

7.6. (b) Clearly $f(x) \geq 0$, and

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \int_0^1 \frac{x}{2} dx + \frac{1}{4} \int_1^2 (3-x) dx + \frac{1}{4} \int_2^3 dx + \frac{1}{4} \int_3^4 (4-x) dx \\ &= \left[\frac{x^2}{4} \right]_0^1 + \frac{1}{4} \left[3x - \frac{x^2}{2} \right]_1^2 + \frac{1}{4} [x]_2^3 + \frac{1}{4} \left[4x - \frac{x^2}{2} \right]_3^4 \\ &= \frac{1}{4} + \frac{1}{4} \left[(6-2) - \left(3 - \frac{1}{2} \right) \right] + \frac{1}{4} [3-2] \\ &\quad + \frac{1}{4} \left[\left(16 - \frac{16}{2} \right) - \left(12 - \frac{9}{2} \right) \right] \\ &= \frac{1}{4} + \frac{1}{4} \left(\frac{3}{2} \right) + \frac{1}{4} + \frac{1}{4} \left(\frac{1}{2} \right) \\ &= \frac{1}{4} + \frac{3}{8} + \frac{1}{4} + \frac{1}{8} = 1. \end{aligned}$$

As $f(x)$ is a density function, therefore

$$\begin{aligned} P(X \geq 3) &= \frac{1}{4} \int_3^4 (4-x) dx = \frac{1}{4} \left[4x - \frac{x^2}{2} \right]_3^4 \\ &= \frac{1}{4} \left[\left(16 - \frac{16}{2} \right) - \left(12 - \frac{9}{2} \right) \right] = \frac{1}{8}; \end{aligned}$$

$P(X=2) = 0$, because for a continuous r.v., the probability at any particular value is equal to zero.

$$P(|X| < 1.5) = P(-1.5 < X < 1.5)$$

$$= \int_{-1.5}^0 0 \cdot dx + \int_0^1 \frac{x}{2} dx + \frac{1}{4} \int_1^{1.5} (3-x) dx$$

$$= \left[\frac{x^2}{4} \right]_0^1 + \frac{1}{4} \left[3x - \frac{x^2}{2} \right]_1^{1.5} = \frac{1}{4} + \frac{1}{4} \left[\left(\frac{9}{2} - \frac{9}{8} \right) - \left(3 - \frac{1}{2} \right) \right]$$

$$= \frac{1}{4} + \frac{1}{4} \left[\frac{27}{8} - \frac{5}{2} \right] = \frac{1}{4} + \frac{7}{32} = \frac{15}{32}, \text{ and}$$

$$P(1 < X < 3) = \frac{1}{4} \int_1^2 (3-x) dx + \frac{1}{4} \int_2^3 dx$$

$$= \frac{1}{4} \left[3x - \frac{x^2}{2} \right]_1^2 + \frac{1}{4} [x]_2^3$$

$$= \frac{1}{4} \left[\left(6 - \frac{4}{2} \right) - \left(3 - \frac{1}{2} \right) \right] + \frac{1}{4} [3 - 2]$$

$$= \frac{1}{4} \left(4 - \frac{5}{2} \right) + \frac{1}{4} = \frac{3}{8} + \frac{1}{4} = \frac{5}{8}$$

7.7. (b) Given $f(x) = A(2-x)(2+x)$, if $0 \leq x \leq 2$
 $= 0$, elsewhere

(i) To find the value of A , we must have that

$$\int_0^2 A(2-x)(2+x) dx = 1$$

$$\text{Solving, } A \int_0^2 (4 - x^2) dx = 1 \text{ or } A \left[4x - \frac{x^3}{3} \right]_0^2 = 1 \text{ which}$$

$$\text{gives } A = \frac{3}{16}.$$

(ii) $P\left(X = \frac{1}{2}\right) = 0$, as probability at a particular value is zero.

$$\begin{aligned} \text{(iii)} \quad P(X \leq 1) &= \frac{3}{16} \int_0^1 (4 - x^2) dx = \frac{3}{16} \left[4x - \frac{x^3}{3} \right]_0^1 \\ &= \frac{3}{16} \left[\left(4 - \frac{1}{3} \right) - 0 \right] = \frac{11}{16}. \end{aligned}$$

(iv) $P(X \geq 2) = 0$, as outside the range 0 to 2, $f(x) = 0$.

$$\begin{aligned} \text{(v)} \quad P(1 \leq X \leq 2) &= \frac{3}{16} \int_1^2 (4 - x^2) dx = \frac{3}{16} \left[4x - \frac{x^3}{3} \right]_1^2 \\ &= \frac{3}{16} \left[\left(8 - \frac{8}{3} \right) - \left(4 - \frac{1}{3} \right) \right] \\ &= \frac{3}{16} \left[\frac{16}{3} - \frac{11}{3} \right] = \frac{3}{16} \times \frac{5}{3} = \frac{5}{16}. \end{aligned}$$

7.8. (i) The function $f(x)$ will be a density function, if

$$f(x) \geq 0 \text{ for every } x, \text{ and } \int_{-\infty}^{\infty} f(x) dx = 1.$$

Now the first condition is obvious and the second condition will be satisfied if

$$\int_0^1 6x(1-x) dx = 1,$$

$$\text{i.e. } 6 \int_0^1 (x - x^2) dx = 1 \quad \text{i.e. } 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = 1$$

$$\text{i.e. } 6 \left[\frac{1}{2} - \frac{1}{3} \right] = 1, \text{ which is true.}$$

Therefore $f(x)$ is a legitimate *p.d.f.*

(ii) The cumulative distribution function (*c.d.f.*) is obtained as

$$\begin{aligned} F(x) &= \int_{-\infty}^x f(x) dx = 6 \int_0^x x(1-x) dx \\ &= 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^x = 3x^2 - 2x^3 \end{aligned}$$

(iii) Now the probability in the interval $\frac{1}{3}$ to $\frac{2}{3}$ is

$$\begin{aligned}
 P\left(\frac{1}{3} < X < \frac{2}{3}\right) &= \int_{1/3}^{2/3} f(x) dx \\
 &= 6 \int_{1/3}^{2/3} (x - x^2) dx = 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_{1/3}^{2/3} \\
 &= 6 \left[\left(\frac{1}{2} \cdot \frac{4}{9} - \frac{1}{3} \cdot \frac{8}{27} \right) - \left(\frac{1}{2} \cdot \frac{1}{9} - \frac{1}{3} \cdot \frac{1}{27} \right) \right] \\
 &= 6 \left[\frac{10}{81} - \frac{7}{162} \right] = 6 \times \frac{13}{162} = \frac{13}{27}; \text{ and}
 \end{aligned}$$

$$P\left(X \leq \frac{1}{2} / \frac{1}{3} \leq X \leq \frac{2}{3}\right) = \frac{P\left(\frac{1}{3} \leq X \leq \frac{1}{2}\right)}{P\left(\frac{1}{3} \leq X \leq \frac{2}{3}\right)}, \text{ where}$$

$$\begin{aligned}
 P\left(\frac{1}{3} \leq X \leq \frac{1}{2}\right) &= \int_{1/3}^{1/2} 6(x - x^2) dx = 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_{1/3}^{1/2} \\
 &= 6 \left[\left(\frac{1}{2} \cdot \frac{1}{4} - \frac{1}{3} \cdot \frac{1}{8} \right) - \left(\frac{1}{2} \cdot \frac{1}{9} - \frac{1}{3} \cdot \frac{1}{27} \right) \right] \\
 &= 6 \left[\frac{1}{12} - \frac{7}{162} \right] = 6 \times \frac{13}{324} = \frac{13}{54},
 \end{aligned}$$

$$\therefore P\left(X \leq \frac{1}{2} / \frac{1}{3} \leq X \leq \frac{2}{3}\right) = \left(\frac{13}{54}\right) / \left(\frac{13}{27}\right) = \frac{13}{54} \times \frac{27}{13} = \frac{1}{2}.$$

(iv) Now $P(X < b) = 6 \int_0^b x(1 - x) dx$

$$= 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^b = 3b^2 - 2b^3; \text{ and}$$

$$P(X > b) = 6 \int_b^1 x(1-x) dx$$

$$= 6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_b^1 = 6 \left[\left(\frac{1}{2} - \frac{1}{3} \right) - \left(\frac{b^2}{2} - \frac{b^3}{3} \right) \right]$$

$$= 1 - 3b^2 + 2b^3$$

We are given that $P(X < b) = 2P(X > b)$

$$\text{i.e.} \quad 3b^2 - 2b^3 = 2(1 - 3b^2 + 2b^3)$$

$$\text{i.e.} \quad 6b^3 - 9b^2 = -2$$

$$\text{i.e.} \quad b^3 - \frac{3}{2}b^2 + \frac{1}{3} = 0.$$

Solving this equation by the Newton-Raphson method, we find that $b = 0.6130$.

Note: The Newton-Raphson method and solution is given below:

Newton-Raphson Formula. Let x_r be an approximation to real root of a function $f(x) = 0$. Then a better approximation, x_{r+1} is given by

$$x_{r+1} = x_r - \frac{f(x_r)}{f'(x_r)}$$

That is, if x_0 is the initial root, then

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)},$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}, \text{ etc.}$$

are the first approximation, the second approximation, etc.

Here $f(x) = x^3 - \frac{3}{2}x^2 + \frac{1}{3}$, and (replacing b by x)

$$f'(x) = 3x^2 - 3x.$$

$$\text{Thus } x_{r+1} = x_r - \frac{x_r^3 - \frac{3}{2}x_r^2 + \frac{1}{3}}{3x_r^2 - 3x_r} = \frac{2x_r^3 - \frac{3}{2}x_r^2 - \frac{1}{3}}{3x_r^2 - 3x_r}$$

Let the approximate root be 0.5. Then for $r=0$, and $x_0 = 0.5$, we have

$$x_1 = \frac{2(0.5)^3 - \frac{3}{2}(0.5)^2 - \frac{1}{3}}{3(0.5)^2 - 3(0.5)} = \frac{-0.458}{-0.75} = 0.6107$$

Putting $r = 1$ and $x_1 = 0.6107$, we find $x_2 = 0.6130$ as a better approximation.

7.9. (i) Let A be the event that a tube will last less than 200 hours, and B be the event that the tube functions after 150 hours of service. We seek $P(A/B)$ which is given as

$$\begin{aligned} P(A/B) &= \frac{P(A \cap B)}{P(B)} \\ \text{Now } P(A \cap B) &= \int_{150}^{200} \frac{100}{x^2} dx \\ &= \left[\frac{-100}{x} \right]_{150}^{200} = \frac{100}{150} - \frac{100}{200} = \frac{2}{3} - \frac{1}{2} = \frac{1}{6} \\ P(B) &= \int_{150}^{\infty} \frac{100}{x^2} dx = \left[\frac{-100}{x} \right]_{150}^{\infty} = \frac{100}{150} = \frac{2}{3} \end{aligned}$$

$$\text{Hence } P(A/B) = \frac{1}{6} \div \frac{2}{3} = \frac{1}{6} \times \frac{3}{2} = \frac{1}{4}.$$

(ii) Probability that exactly one out of 3 such tubes will have to be replaced after 150 hours of service (i.e. one not functioning and 2 functioning) is

$$\binom{3}{1} \left(1 - \frac{1}{4}\right) \left(\frac{1}{4}\right)^2 = \frac{9}{64}.$$

(iii) The maximum no. n, of tubes is obtained by solving

$$\left(\frac{2}{3}\right)^n = 0.5, \text{ which gives } n = 2.$$

7.10. (b) The given joint p.d. of two r.v.'s is presented in the following table:

$X \backslash Y$	1	2	3	$P(X=x_i)$ $g(x)$
1	6/30	1/30	1/30	8/30
2	4/30	5/30	1/30	10/30
3	2/30	4/30	6/30	12/30
$P(Y=y_j):h(y)$	$\frac{12}{30}$	$\frac{10}{30}$	$\frac{8}{30}$	1

The marginal p.d. of X is obtained by adding over the rows and that of Y , by adding over the columns. Thus the two marginal distributions are:

x	1	2	3
$g(x)$	$\frac{8}{30}$	$\frac{10}{30}$	$\frac{12}{30}$

and

y	1	2	3
$h(y)$	$\frac{12}{30}$	$\frac{10}{30}$	$\frac{8}{30}$

By definition the conditional p.d. of $X = x_i$ given $Y = y_j$ is

$$f(x_i/y_j) = P(X=x_i/Y=y_j) = \frac{f(x_i, y_j)}{h(y_j)}, \text{ for } i = 1, 2, 3; j = 1, 2, 3.$$

Thus $f(1/1) = \frac{f(1, 1)}{h(1)} = \frac{6/30}{12/30} = \frac{1}{2};$

$$f(2/1) = \frac{f(2, 1)}{h(1)} = \frac{4/30}{12/30} = \frac{1}{3};$$

$$f(3/1) = \frac{f(3, 1)}{h(1)} = \frac{2/30}{12/30} = \frac{1}{6}.$$

Hence the conditional p.d. of X given that $Y = 1$ is

x	1	2	3
$f(x/1)$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{6}$

Similarly the conditional distributions of X given that $y = 2, 3$ could be found.

The conditional p.d. of Y for given $X = x_i$ is

$$f(y_j/x_i) = P(Y=y_j/X=x_i) = \frac{f(x_i/y_j)}{g(x_i)}.$$

Thus $f(Y/2) = \frac{f(2, y)}{g(2)}$ for $y = 1, 2, 3$. That is

$$f(1/2) = \frac{f(2, 1)}{g(2)} = \frac{4/30}{10/30} = \frac{2}{5};$$

$$f(2/2) = \frac{f(2, 2)}{g(2)} = \frac{5/30}{10/30} = \frac{1}{2}; \text{ and}$$

$$f(3/2) = \frac{f(2, 3)}{g(2)} = \frac{1/30}{10/30} = \frac{1}{10}.$$

Hence the conditional *p.d.* of Y given that $X = 2$ is

y	1	2	3
$f(y/2)$	$\frac{2}{5}$	$\frac{1}{2}$	$\frac{1}{10}$

Similarly the conditional distributions of Y given that $X=1$ and 3 could be found.

7.11. The marginal distributions $g(x)$ and $h(y)$ are obtained by adding over the columns and rows respectively in the following table:

$Y \backslash X$	1	2	3	$h(y)$
1	$1/12$	$1/6$	0	$3/12$
2	0	$1/9$	$1/5$	$14/45$
3	$1/18$	$1/4$	$2/15$	$79/180$
$g(x)$	$\frac{5}{36}$	$\frac{19}{36}$	$\frac{5}{15}$	1

Thus the marginal distributions are:

x	1	2	3
$g(x)$	$\frac{5}{36}$	$\frac{19}{36}$	$\frac{5}{15}$

and

y	1	2	3
$h(y)$	$\frac{3}{12}$	$\frac{14}{45}$	$\frac{79}{180}$

By definition, the conditional *p.d.* of $X = x_i$ given $Y = y_j$ is

$$f(x_i/y_j) = P(X=x_i/Y=y_j) = \frac{f(x_i, y_j)}{h(y_j)}$$

For $y = 1$, the conditional distribution of X is

$$f(1/1) = \frac{f(1, 1)}{h(1)} = \frac{1/12}{3/12} = \frac{1}{3};$$

$$f(2/1) = \frac{f(2, 1)}{h(1)} = \frac{1/6}{3/12} = \frac{2}{3}; \text{ and}$$

$$f(3/1) = \frac{f(3, 1)}{h(1)} = \frac{0}{3/12} = 0.$$

Hence the conditional *p.d.* of X given that $Y = 1$ is

x	1	2	3
$f(x/1)$	$\frac{1}{3}$	$\frac{2}{3}$	0

Similarly the conditional distributions of X could be found for given $y = 2, 3$ respectively.

The conditional *p.d.* of $Y = y_j$ for given $X = x_i$ are

$$f(y_j/x_i) = P(Y=y_j/X=x) = \frac{f(x, y)}{g(x)}$$

Thus for $x = 3$, the conditional distribution of Y is

$$f(1/3) = \frac{f(3, 1)}{g(3)} = \frac{0}{5/15} = 0;$$

$$f(2/3) = \frac{f(3, 2)}{g(3)} = \frac{1/5}{5/15} = \frac{3}{5}; \text{ and}$$

$$f(3/3) = \frac{f(3, 3)}{g(3)} = \frac{2/15}{5/15} = \frac{2}{5}$$

Hence the conditional distribution of Y given that $X = 3$ is

y	1	2	3
$f(y/3)$	0	$\frac{3}{5}$	$\frac{2}{5}$

Similarly for $x=1$ and $x=2$, conditional p.d. of y could be found.

(b) The r.v.'s X and Y will be independent, if

$$f(x, y) = g(x) \cdot h(y).$$

Now $f(1, 1) = \frac{1}{12}$ and $g(1) \cdot h(1) = \left(\frac{3}{12}\right) \left(\frac{5}{36}\right) = \frac{5}{144}$.

Since $f(1, 1) \neq g(1) \cdot h(1)$, therefore X and Y are not independent.

7.12. (i) The marginal probability distribution for X is

$$\begin{aligned} g(x) &= \sum_y f(x, y) \\ &= \sum_{y=1}^3 \frac{xy}{66} = \frac{x}{66} + \frac{2x}{66} + \frac{3x}{66} = \frac{x}{11}, \text{ for } x=2,4,5; \end{aligned}$$

and the marginal probability distribution for Y is

$$\begin{aligned} h(y) &= \sum_x f(x, y) \\ &= \sum_x \frac{xy}{66} = \frac{2y}{66} + \frac{4y}{66} + \frac{5y}{66} = \frac{y}{6}, \text{ for } y=1,2,3. \end{aligned}$$

Now $\frac{xy}{66} = \frac{x}{11} \times \frac{y}{6}$.

i.e. $f(x, y) = g(x)h(y)$, therefore X and Y are independent.

(ii) The marginal p.d. for X is

$$\begin{aligned} g(x) &= \sum_y f(x, y) \\ &= \sum_{y=1}^2 \frac{xy^2}{30} = \frac{x}{30} + \frac{4x}{30} = \frac{x}{6}, \text{ for } x=1,2,3; \text{ and} \end{aligned}$$

the marginal p.d. for Y is

$$\begin{aligned} h(y) &= \sum_x f(x, y) \\ &= \sum_{x=1}^2 \frac{xy^2}{30} = \frac{y^2}{30} + \frac{2y^2}{30} + \frac{3y^2}{30} = \frac{y^2}{5}, \text{ for } y=1,2. \end{aligned}$$

Now $\frac{xy^2}{30} = \frac{x}{6} \times \frac{y^2}{5}$, i.e. $f(x, y) = g(x) \cdot h(y)$

Hence X and Y are independent.

7.13. (b) Given the joint p.d.f. $f(x,y) = 3xy(x+y)$, for $0 \leq x \leq 1$,
 $0 \leq y \leq 1$.

The marginal p.d.f. of X is

$$\begin{aligned} g(x) &= 3 \int_0^1 xy(x+y) dy, & 0 \leq x \leq 1, \\ &= 3 \int_0^1 (x^2y + xy^2) dy, & 0 \leq x \leq 1, \\ &= 3 \left[\frac{x^2y^2}{2} + \frac{xy^3}{3} \right]_0^1 & 0 \leq x \leq 1, \\ &= 3 \left[\frac{x^2}{2} + \frac{x}{3} \right] & 0 \leq x \leq 1, \\ &= \frac{3x^2}{2} + x, & 0 \leq x \leq 1. \end{aligned}$$

Similarly, the marginal p.d.f. of Y is

$$\begin{aligned} h(y) &= 3 \int_0^1 (x^2y + xy^2) dx, & 0 \leq y \leq 1, \\ &= 3 \left[\frac{x^3y}{3} + \frac{x^2y^2}{2} \right]_0^1 & 0 \leq y \leq 1, \\ &= 3 \left[\frac{y}{2} + \frac{y^2}{2} \right] & 0 \leq y \leq 1, \\ &= y + \frac{3}{2}y^2. & 0 \leq y \leq 1. \end{aligned}$$

The conditional p.d.f. of X given $Y=y$ is

$$\begin{aligned} f(x/y) &= \frac{f(x,y)}{h(y)}, \text{ where } h(y) > 0 \\ &= \frac{3xy(x+y)}{y + \frac{3}{2}y^2} = \frac{3x(x+y)}{1 + \frac{3}{2}y}, \end{aligned}$$

and the conditional *p.d.f.* of *Y* given *X* = *x*, is

$$f(y/x) = \frac{f(x, y)}{g(y)} = \frac{3xy (x + y)}{x (1 + \frac{3}{2}x)} = \frac{3y (x + y)}{1 + \frac{3}{2}x}.$$

7.14. Given that

$$f(x, y) = x^2 + \frac{1}{3}xy, \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 2$$

$$\begin{aligned} \text{(a)} \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy &= \int_0^1 \int_0^2 (x^2 + \frac{1}{3}xy) dy dx \\ &= \int_0^1 \left[x^2y + \frac{1}{6}xy^2 \right]_0^2 dx \\ &= \int_0^1 \left(2x^2 + \frac{2x}{3} \right) dx \\ &= \left[\frac{2x^3}{3} + \frac{x^2}{3} \right]_0^1 = \frac{2}{3} + \frac{1}{3} = 1. \end{aligned}$$

$$\begin{aligned} \text{(b) (i)} \quad P(X > \frac{1}{2}) &= \int_{1/2}^1 \int_0^2 (x^2 + \frac{1}{3}xy) dy dx \\ &= \int_{1/2}^1 \left[x^2y + \frac{xy^2}{6} \right]_0^2 dx = \int_{1/2}^1 \left(2x^2 + \frac{2x}{3} \right) dx \\ &= \left[\frac{2x^3}{3} + \frac{x^2}{3} \right]_{1/2}^1 = \left(\frac{2}{3} + \frac{1}{3} \right) - \left(\frac{1}{12} + \frac{1}{12} \right) \\ &= 1 - \frac{2}{12} = \frac{5}{6}. \end{aligned}$$

$$\text{(ii)} P(Y < X) = \int_0^1 \int_0^x (x^2 + \frac{1}{3}xy) dy dx \quad (\because y < x)$$

$$= \int_0^1 \left[x^2 y + \frac{1}{6} x y^2 \right]_0^x dx = \int_0^1 \left(x^3 + \frac{x^3}{6} \right) dx$$

$$= \frac{7}{6} \int_0^1 x^3 dx = \frac{7}{6} \left[\frac{x^4}{4} \right]_0^1 = \frac{7}{6} \times \frac{1}{4} = \frac{7}{24}.$$

$$(iii) \quad P(X + Y > 1) = 1 - P(X + Y < 1)$$

$$= 1 - \int_0^1 \int_0^{1-x} \left(x^2 + \frac{1}{3} xy \right) dy dx$$

$$(\because x+y < 1, \therefore y = 1-x)$$

$$= 1 - \int_0^1 \left[x^2 y + \frac{1}{6} x y^2 \right]_0^{1-x} dx$$

$$= 1 - \int_0^1 \left\{ x^2 (1-x) + \frac{1}{6} x (1-x)^2 \right\} dx$$

$$= 1 - \int_0^1 \frac{1}{6} (x + 4x^2 - 5x^3) dx$$

$$= 1 - \frac{1}{6} \left[\frac{x^2}{2} + \frac{4x^3}{3} - \frac{5x^4}{4} \right]_0^1$$

$$= 1 - \frac{1}{6} \left[\frac{1}{2} + \frac{4}{3} - \frac{5}{4} \right] = 1 - \frac{1}{6} \left(\frac{7}{12} \right)$$

$$= 1 - \frac{7}{72} = \frac{65}{72}.$$

$$(iv) \quad \text{Now } P\left(Y < \frac{1}{2} \mid X < \frac{1}{2}\right) = \frac{P\left(Y < \frac{1}{2} \text{ and } X < \frac{1}{2}\right)}{P\left(X < \frac{1}{2}\right)}, \text{ and}$$

$$\begin{aligned}
 P(Y < \frac{1}{2} \text{ and } X < \frac{1}{2}) &= \int_0^{1/2} \int_0^{1/2} (x^2 + \frac{1}{3}xy) dy dx \\
 &= \int_0^{1/2} \left[x^2y + \frac{1}{6}xy^2 \right]_0^{1/2} dx \\
 &= \int_0^{1/2} \left(\frac{x^2}{2} + \frac{x}{24} \right) dx \\
 &= \left[\frac{x^3}{6} + \frac{x^2}{48} \right]_0^{1/2} = \frac{1}{48} + \frac{1}{192} = \frac{5}{192}
 \end{aligned}$$

$$\therefore P(Y < \frac{1}{2} | X < \frac{1}{2}) = \frac{5/192}{1/6} = \frac{5}{32}, \quad (\because P(X < \frac{1}{2}) = \frac{1}{6}).$$

7.15.. (b) The marginal density functions are obtained as below:

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy, \quad 0 < x < 1,$$

$$= \frac{2}{3} \int_0^1 (x + 2y) dy, \quad 0 < x < 1,$$

$$= \frac{2}{3} \left[xy + \frac{2y^2}{2} \right]_0^1 = \frac{2}{3} [x + 1] \quad 0 \leq x < 1.$$

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx \quad 0 < y < 1,$$

$$= \frac{2}{3} \int_0^1 (x + 2y) dx, \quad 0 < y < 1,$$

$$= \frac{2}{3} \left[\frac{x^2}{2} + 2xy \right]_0^1, \quad 0 < y < 1,$$

$$= \frac{2}{3} \left[\frac{1}{2} + 2y \right], \quad 0 < y < 1,$$

$$= \frac{1}{3} + \frac{2}{3}y \quad 0 < y < 1.$$

The conditional *p.d.f.* of X given $Y = y$ is

$$f(x/y) = \frac{f(x, y)}{h(y)}, \text{ where } h(y) > 0.$$

$$= \frac{\frac{2}{3}(x + 2y)}{\frac{2}{3} \left[\frac{1}{2} + 2y \right]} = \frac{x + 2y}{\frac{1}{2} + 2y} = \frac{2x + 4y}{1 + 4y};$$

and the conditional *p.d.f.* of Y given $X = x$ is

$$f(y/x) = \frac{f(x, y)}{g(x)}, \text{ where } g(x) > 0$$

$$= \frac{\frac{2}{3}(x + 2y)}{\frac{2}{3}(x + 1)} = \frac{x + 2y}{x + 1}.$$

We know that $f(x/y) = \frac{2x + 4y}{1 + 4y}$, therefore

$$P(X < \frac{1}{2} / y = \frac{1}{2}) = \int_0^{1/2} f(x/\frac{1}{2}) dx = \int_0^{1/2} \frac{2x + 4(\frac{1}{2})}{1 + 4(\frac{1}{2})} dx$$

$$= \int_0^{1/2} \frac{2x + 2}{3} \cdot dx = \frac{2}{3} \left[\frac{x^2}{2} + x \right]_0^{1/2}$$

$$= \frac{2}{3} \left[\frac{1}{8} + \frac{1}{2} \right] = \frac{2}{3} \times \frac{5}{8} = \frac{5}{12}.$$

7.16. The marginal density functions are obtained below:

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$= 3 \int_0^1 (x^2y + y^2x) dy = 3 \left[\frac{x^2y^2}{2} + \frac{y^3x}{3} \right]_0^1$$

$$= \frac{3}{2}x^2 + x, \quad 0 \leq x \leq 1.$$

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

$$= 3 \int_0^1 (x^2y + y^2x) dx = 3 \left[\frac{x^3y}{3} + \frac{y^2x^2}{2} \right]_0^1$$

$$= y + \frac{3}{2}y^2. \quad 0 \leq y \leq 1.$$

The conditional distributions are

$$f(x/y) = \frac{f(x, y)}{h(y)} = \frac{3xy(x + y)}{y + (3/2)y^2} = \frac{6x(x + y)}{2 + 3y}, \text{ and}$$

$$f(y/x) = \frac{f(x, y)}{g(x)} = \frac{3xy(x + y)}{x + (3/2)x^2} = \frac{6y(x + y)}{2 + 3x}.$$

The conditional probability is obtained below:

$$P\left[\frac{1}{2} \leq x \leq \frac{3}{4} \mid \frac{1}{2} \leq y \leq \frac{2}{3}\right] = \frac{\int_{1/3}^{3/4} \int_{1/2}^{2/3} (3x^2y + 3y^2x) dy dx}{\int_{1/2}^{2/3} (y + \frac{3}{2}y^2) dy}$$

Now

$$\int_{1/2}^{3/4} \int_{1/2}^{2/3} (3x^2y + 3y^2x) dy dx = \int_{1/2}^{3/4} \left[\frac{3x^2y^2}{2} + \frac{3y^2x}{3} \right]_{1/2}^{2/3} dx$$

$$= \int_{1/2}^{3/4} \left(\frac{7}{24}x^2 + \frac{37}{216}x \right) dx$$

$$= \left[\frac{7}{24} \cdot \frac{x^3}{3} + \frac{37}{216} \cdot \frac{x^2}{2} \right]_{1/2}^{3/4} = \frac{769}{13824}$$

$$\text{and } \int_{1/2}^{2/3} \left(y + \frac{3}{2} y^2 \right) dy = \left[\frac{y^2}{2} + \frac{3y^3}{6} \right]_{1/2}^{2/3} = \frac{79}{432}$$

Hence the required conditional probability is

$$\frac{769}{13824} \times \frac{432}{79} = \frac{769}{2528} = 0.304.$$

7.17. The marginal distribution of X is obtained as

$$\begin{aligned} g(x) &= \int_{-\infty}^{\infty} f(x, y) dy \\ &= 24 \int_0^1 x^2 y (1-x) dy = 24 \left[\frac{x^2 y^2 (1-x)}{2} \right]_0^1 \\ &= 12x^2 (1-x) \end{aligned}$$

The conditional distribution of Y given X is

$$f(y/x) = \frac{f(x, y)}{g(x)} = \frac{24 x^2 y (1-x)}{12x^2 (1-x)} = 2y$$

Also the marginal distribution of Y is obtained as

$$\begin{aligned} h(y) &= \int_{-\infty}^{\infty} f(x, y) dx \\ &= 24 \int_0^1 x^2 y (1-x) dx = 24 \left[y \left(\frac{x^3}{3} - \frac{x^4}{4} \right) \right]_0^1 = 2y \end{aligned}$$

Now obviously $f(x, y) = g(x) \cdot h(y)$

Hence X and Y are independent.

7.19. (c) Let the r.v. X represent the sum of the spots on the dice. Then X has the following p.d.

x	2	3	4	5	6	7	8	9	10	11	12
$f(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

The expected value of the payment is

$$E(X) = 2 \times \frac{1}{36} + 3 \times \frac{2}{36} + 4 \times \frac{3}{36} + \dots + 12 \times \frac{1}{36} = \frac{252}{36} = 7$$

7.20. (a) Let the r.v. X represent the number of women on the committee. Then X has the following p.d.

x	$f(x)$	$xf(x)$
0	$\binom{3}{0} \binom{5}{5} \div \binom{8}{5} = \frac{1}{56}$	0
1	$\binom{3}{1} \binom{5}{4} \div \binom{8}{5} = \frac{15}{56}$	$\frac{15}{56}$
2	$\binom{3}{2} \binom{5}{3} \div \binom{8}{5} = \frac{30}{56}$	$\frac{60}{56}$
3	$\binom{3}{3} \binom{5}{2} \div \binom{8}{5} = \frac{10}{56}$	$\frac{30}{56}$
Σ	1	$\frac{105}{56}$

Hence the expected number of women on the committee is

$$E(X) = \sum x f(x) = \frac{105}{56}$$

(b) The mathematical expectation of a random variable X , i.e. $E(X)$ is defined as

$$E(X) = \sum_i x_i f(x_i)$$

the summation extending over all possible values of X , provided that $\sum_i |x_i| f(x_i)$ converges.

Now $x_i = 2, -2, \frac{8}{3}, -4, \dots, (-1)^{j+1} \frac{2j}{j}, \dots$

$$f(x_i) = \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots, \left(\frac{1}{2}\right)^j, \dots$$

$$\therefore \sum_j |x_j| f(x_j) = \sum_j \frac{2j}{j} \left(\frac{1}{2}\right)^j$$

$$= \sum_{j=1}^{\infty} \frac{1}{j},$$

which is a divergent series, and hence $E(X)$ does not exist. It is however interesting to note that

$$\sum_j x_j f(x_j) = \sum_{j=1}^{\infty} (-1)^{j+1} \frac{1}{j}, \text{ converges.}$$

7.21. (b) As the variables X and Y are independent, therefore their joint distribution is

$\begin{matrix} X \\ Y \end{matrix}$	1	2	3	4	$f(y)$	$yf(y)$
0	1/32	1/32	1/32	1/32	1/8	0
1	3/32	3/32	3/32	3/32	3/8	3/8
2	3/32	3/32	3/32	3/32	3/8	6/8
3	1/32	1/32	1/32	1/32	1/8	3/8
$f(x)$	1/4	1/4	1/4	1/4	1	12/8
$xf(x)$	1/4	2/4	3/4	4/4	10/4	

Now $E(X) = \sum xf(x) = \frac{10}{4} = 2.5$; and

$$E(Y) = \sum yf(y) = \frac{12}{4} = 1.5;$$

Again $E(X+Y) = \sum (x+y) f(x, y)$

$$\begin{aligned}
 &= \left(1 \times \frac{1}{32}\right) + \left(2 \times \frac{1}{32}\right) + \left(3 \times \frac{1}{32}\right) + \left(4 \times \frac{1}{32}\right) \\
 &\quad + \left(2 \times \frac{3}{32}\right) + \left(3 \times \frac{3}{32}\right) + \left(4 \times \frac{3}{32}\right) + \left(5 \times \frac{3}{32}\right) \\
 &\quad + \left(3 \times \frac{3}{32}\right) + \left(4 \times \frac{3}{32}\right) + \left(5 \times \frac{3}{32}\right) + \left(6 \times \frac{3}{32}\right) \\
 &\quad + \left(4 \times \frac{1}{32}\right) + \left(5 \times \frac{1}{32}\right) + \left(6 \times \frac{1}{32}\right) + \left(7 \times \frac{1}{32}\right) \\
 &= \frac{128}{32} = 4
 \end{aligned}$$

Also $E(X) + E(Y) = 2.5 + 1.5 = 4$

Hence $E(X) + E(Y) = E(X + Y)$

7.22 (a) Computation of the expected values $E(X)$ and $E(X^2)$.

x_i	$f(x_i)$	$x_i f(x_i)$	$x_i^2 f(x_i)$
0	$\binom{3}{0} \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^3 = \frac{27}{64}$	0	0
1	$\binom{3}{1} \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^2 = \frac{27}{64}$	$\frac{27}{64}$	$\frac{27}{64}$
2	$\binom{3}{2} \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^1 = \frac{9}{64}$	$\frac{18}{64}$	$\frac{36}{64}$
3	$\binom{3}{3} \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^0 = \frac{1}{64}$	$\frac{3}{64}$	$\frac{9}{64}$
Σ	1	$\frac{48}{64}$	$\frac{72}{64}$

Hence $E(X) = \Sigma x_i f(x_i) = \frac{48}{64} = \frac{3}{4}$, and

$$E(X^2) = \Sigma x_i^2 f(x_i) = \frac{72}{64} = 1 \frac{1}{8}.$$

(b) (i) Computation of $E(X)$ and $\text{Var}(X)$.

x_i	$f(x_i)$	$x_i f(x_i)$	$x_i^2 f(x_i)$
-1	0.125	-0.125	0.125
0	0.500	0.000	0.000
1	0.200	0.200	0.200
2	0.050	0.100	0.200
3	0.125	0.375	1.125
Total	1.000	0.550	1.650

Thus $E(X) = \Sigma x f(x) = 0.55$; and

$$\begin{aligned} \text{Var}(X) &= E(X^2) - [E(X)]^2 = 1.650 - (0.55)^2 = 1.650 - 0.3025 \\ &= 1.3475 = 1.35 \end{aligned}$$

(ii) Computation of the p.d. of $Y = 2X + 1$, $E(Y)$ and $\text{Var}(Y)$.

$y = (2x + 1)$	$f(y)$	$yf(y)$	$y^2f(y)$
-1	0.125	-0.125	0.125
1	0.500	0.500	0.500
3	0.200	0.600	1.800
5	0.050	0.250	1.250
7	0.125	0.875	6.125
Total	1.000	2.100	9.800

$$\therefore E(Y) = \sum yf(y) = 2.1, \text{ and}$$

$$\text{Var}(Y) = E(Y^2) - [E(Y)]^2 = 9.80 - (2.1)^2 = 9.80 - 4.41 = 5.39.$$

(iii) For the relationship between $E(X)$ and $E(Y)$ where $Y = 2X + 1$, we have

$$\begin{aligned} E(Y) &= 2.1 \\ &= 2(0.55) + 1 = 2E(X) + 1. \end{aligned}$$

For the relationship between $\text{Var}(X)$ and $\text{Var}(Y)$, we have

$$\begin{aligned} \text{Var}(Y) &= 5.39 \\ &= 4(1.3475) = (2)^2 \text{Var}(X). \end{aligned}$$

7.23 (a) The function $f(x)$ will be a density function, if

$$A \int_0^1 x^3(1-x) dx = 1, \text{ i.e. if } A \int_0^1 (x^3 - x^4) dx = 1,$$

$$\text{i.e. if } A \left[\frac{x^4}{4} - \frac{x^5}{5} \right]_0^1 = 1, \text{ i.e. if } A = 20.$$

$f(x) = 20x^3(1-x)$ is a proper density function.

(b) The mean and variance are calculated below:

$$\text{Now } E(X) = \int_{-\infty}^{\infty} xf(x) dx$$

$$= 20 \int_0^1 x(x^3 - x^4) dx = 20 \int_0^1 (x^4 - x^5) dx$$

$$= 20 \left[\frac{x^5}{5} - \frac{x^6}{6} \right]_0^1 = 20 \left[\frac{1}{5} - \frac{1}{6} \right] = \frac{2}{3},$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$= 20 \int_0^1 x^2(x^3 - x^4) dx = 20 \int_0^1 (x^5 - x^6) dx$$

$$= 20 \left[\frac{x^6}{6} - \frac{x^7}{7} \right]_0^1 = 20 \left[\frac{1}{6} - \frac{1}{7} \right] = \frac{10}{21}$$

$$\therefore \text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{10}{21} - \left(\frac{2}{3}\right)^2 = \frac{2}{63}$$

(c) The distribution function is

$$F(x) = P(X \leq x) = 20 \int_0^x x^3(1-x) dx = 20 \int_0^x (x^3 - x^4) dx$$

$$= 20 \left[\frac{x^4}{4} - \frac{x^5}{5} \right]_0^x = 20 \left[\frac{x^4}{4} - \frac{x^5}{5} \right]$$

$$= 5x^4 - 4x^5$$

$$\text{Now } P\left(\frac{1}{4} < X < \frac{1}{2}\right) = P\left(X < \frac{1}{2}\right) - P\left(X < \frac{1}{4}\right)$$

$$= F\left(\frac{1}{2}\right) - F\left(\frac{1}{4}\right)$$

$$= \left[5\left(\frac{1}{2}\right)^4 - 4\left(\frac{1}{2}\right)^5 \right] - \left[5\left(\frac{1}{4}\right)^4 - 4\left(\frac{1}{4}\right)^5 \right]$$

$$= \left(\frac{5}{16} - \frac{4}{32} \right) - \left(\frac{5}{256} - \frac{4}{1024} \right)$$

$$= \frac{6}{32} - \frac{16}{1024} = \frac{3}{16} - \frac{1}{64} = \frac{11}{64}$$

$$\begin{aligned}
 7.24. (a) \quad E[X-E(X)]^2 &= E[X^2 - 2XE(X) + \{E(X)\}^2] \\
 &= E(X^2) - 2E(X) \cdot E(X) + [E(X)]^2 \\
 &= E(X^2) - 2[E(X)]^2 + [E(X)]^2 \\
 &= E(X^2) - [E(X)]^2.
 \end{aligned}$$

(b) Given that $\text{Var}(X_1) = k$, $\text{Var}(X_2) = 2$, and $\text{Var}(3X_2 - X_1) = 25$.

Since X_1 and X_2 are independent r.v.'s, therefore

$$\begin{aligned}
 \text{Var}(3X_2 - X_1) &= 9 \text{Var}(X_2) + \text{Var}(X_1) \\
 &= 9(2) + k
 \end{aligned}$$

Thus $9(2) + k = 25$

or $k = 25 - 18 = 7$.

(c) (i) The probability distribution of r.v. X is

x	1	2	3	4
$f(x)$	$1/8$	$2/8$	$3/8$	$2/8$

(ii) $E(X) = \sum x f(x)$

$$\begin{aligned}
 &= \left(1 \times \frac{1}{8}\right) + \left(2 \times \frac{2}{8}\right) + \left(3 \times \frac{3}{8}\right) + \left(4 \times \frac{2}{8}\right) \\
 &= \frac{22}{8} = 2.75.
 \end{aligned}$$

$\text{Var}(X) = E(X^2) - [E(X)]^2$, where

$E(X^2) = \sum x^2 f(x)$

$$\begin{aligned}
 &= \left(1^2 \times \frac{1}{8}\right) + \left(2^2 \times \frac{2}{8}\right) + \left(3^2 \times \frac{3}{8}\right) + \left(4^2 \times \frac{2}{8}\right) \\
 &= \frac{68}{8}
 \end{aligned}$$

$\therefore \text{Var}(X) = \left(\frac{68}{8}\right) - \left(\frac{22}{8}\right)^2 = \frac{60}{64} = \frac{15}{16}$

7.25. (a) Calculation of mean and variance.

x_i	$f(x_i) = \frac{6 - 7-x }{36}$	$x_i f(x_i)$	$x_i^2 f(x_i)$
2	1/36	2/36	4/36
3	2/36	6/36	18/36
4	3/36	12/36	48/36
5	4/36	20/36	100/36
6	5/36	30/36	180/36
7	6/36	42/36	294/36
8	5/36	40/36	320/36
9	4/36	36/36	324/36
10	3/36	30/36	300/36
11	2/36	22/36	242/36
12	1/36	12/36	144/36
Σ	1	252/36 = 7	1974/36

$$\therefore \text{Mean} = E(X) = \sum x_i f(x_i) = 7, \text{ and}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= \sum x_i^2 f(x_i) - [\sum x_i f(x_i)]^2$$

$$= \frac{1974}{36} - (7)^2 = \frac{210}{36} = \frac{35}{6}.$$

(b) Here $x_i = 1, 2, 3, \dots, n$

$$f(x_i) = \frac{1}{n}, \frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n}$$

$$\therefore E(X) = \sum x_i f(x)$$

$$= 1 \cdot \frac{1}{n} + 2 \cdot \frac{1}{n} + 3 \cdot \frac{1}{n} + \dots + n \cdot \frac{1}{n}$$

$$= \frac{1}{n} [1 + 2 + 3 + \dots + n]$$

$$= \frac{1}{n} \left[\frac{n(n+1)}{2} \right] = \frac{n+1}{2}.$$

$$\begin{aligned}
 E(X^2) &= 1^2 \cdot \frac{1}{n} + 2^2 \cdot \frac{1}{n} + 3^2 \cdot \frac{1}{n} + \dots + n^2 \cdot \frac{1}{n} \\
 &= \frac{1}{n} [1^2 + 2^2 + 3^2 + \dots + n^2] \\
 &= \frac{1}{n} \left[\frac{n(n+1)(2n+1)}{6} \right] = \frac{(n+1)(2n+1)}{6}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Var}(X) &= E(X^2) - [E(X)]^2 \\
 &= \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2} \right)^2 \\
 &= \frac{2n^2 + 3n + 1}{6} - \frac{n^2 + 2n + 1}{4} = \frac{n^2 - 1}{12}
 \end{aligned}$$

(c) Here $x_i = 0, 1, 2, 3, \dots, n$

$$f(x_i) = k \binom{n}{0}, k \binom{n}{1}, k \binom{n}{2}, \dots, k \binom{n}{n}.$$

where k is the constant of proportionality.

Since $\sum f(x_i) = 1$, therefore we first find the value of k , the constant of proportionality. The sum is

$$k \binom{n}{0} + k \binom{n}{1} + k \binom{n}{2} + \dots + k \binom{n}{n} = 1$$

$$\text{or } k \left[1 + n + \frac{n(n-1)}{2!} + \dots + 1 \right] = 1$$

$$\text{or } k [1 + 1]^n = 1 \text{ or } k 2^n = 1 \text{ or } k = \frac{1}{2^n}$$

$$\text{Now } E(X) = \sum x_i f(x_i)$$

$$= \frac{1}{2^n} \left[0 \cdot 1 + 1 \cdot n + 2 \cdot \frac{n(n-1)}{2!} + 3 \cdot \frac{n(n-1)(n-2)}{3!} + \dots + n \cdot 1 \right]$$

$$= \frac{n}{2^n} \left[1 + (n-1) + \frac{(n-1)(n-2)}{2!} + \dots + 1 \right]$$

$$= \frac{n}{2^n} [1 + 1]^{n-1} = \frac{n}{2^n} (2)^{n-1} = \frac{n}{2}.$$

$$\begin{aligned}
 E(X^2) &= \frac{1}{2^n} \left[0^2 \cdot 1 - 1^2 \cdot n + 2^2 \cdot \frac{n(n-1)}{2!} + 3^2 \cdot \frac{n(n-1)(n-2)}{3!} + \dots + n^2 \cdot 1 \right] \\
 &= \frac{1}{2^n} \left[n + 2 \cdot \frac{n(n-1)}{1!} + 3 \cdot \frac{n(n-1)(n-2)}{2!} + \dots + n^2 \right] \\
 &= \frac{n}{2^n} \left[\left\{ 1 + (n-1) + \frac{(n-1)(n-2)}{2!} + \dots + 1 \right\} + \right. \\
 &\quad \left. \left\{ (n-1) + \frac{2(n-1)(n-2)}{2!} + \dots + (n-1) \right\} \right] \\
 &= \frac{n}{2^n} [(1+1)^{n-1} + (n-1)(1+1)^{n-2}] \\
 &= \frac{n}{2^n} [2^{n-1} + (n-1)2^{n-2}] = \frac{n}{2} + \frac{n(n-1)}{2^2}
 \end{aligned}$$

Hence $\text{Var}(X) = E(X^2) - [E(X)]^2$

$$= \frac{n}{2} + \frac{n(n-1)}{4} - \left(\frac{n}{2}\right)^2 = \frac{2n + n^2 - n - n^2}{4} = \frac{n}{4}$$

7.26. (a) The probability of throwing a 6 with one die is $\frac{1}{6}$ and that of not throwing a 6 with one die is $\frac{5}{6}$.

As A has the first throw, therefore he can win in the first, third, fifth, ..., throws.

The probability that A wins in the first throw = $\frac{1}{6}$

The probability that A wins in the 3rd throw = $\left(\frac{5}{6}\right)^2 \cdot \frac{1}{6}$, as both A and B have failed once before A's 3rd throw.

The probability that A wins in the 5th throw = $\left(\frac{5}{6}\right)^4 \cdot \frac{1}{6}$ and so on. Since these cases are mutually exclusive, therefore the probability that A wins = $\frac{1}{6} + \left(\frac{5}{6}\right)^2 \cdot \frac{1}{6} + \left(\frac{5}{6}\right)^4 \cdot \frac{1}{6} + \dots$

$$= \frac{1/6}{1 - \left(\frac{5}{6}\right)^2} = \frac{1}{6} \times \frac{36}{11} = \frac{6}{11}.$$

The probability that B wins $= 1 - P(A \text{ wins})$

$$= 1 - \frac{6}{11} = \frac{5}{11}.$$

Hence A 's expectation $= \frac{6}{11} \times \text{Rs. } 11 = \text{Rs. } 6$, and

$$B\text{'s expectation} = \frac{5}{11} \times \text{Rs. } 11 = \text{Rs. } 5.$$

(b) The probability of getting a spade $= \frac{13}{52} = \frac{1}{4}$, and the probability of not getting a spade $= 1 - \frac{1}{4} = \frac{3}{4}$.

Now A can cut a spade in the first, fifth, ninth, ..., drawings, the respective probabilities of which are

$$\frac{1}{4}, \left(\frac{3}{4}\right)^4 \times \frac{1}{4}, \left(\frac{3}{4}\right)^8 \times \frac{1}{4}, \dots$$

Thus the probability that A cuts a spade first

$$= \frac{1}{4} + \left(\frac{3}{4}\right)^4 \cdot \frac{1}{4} + \left(\frac{3}{4}\right)^8 \cdot \frac{1}{4} + \dots$$

$$= \frac{1}{4} \left[1 + \left(\frac{3}{4}\right)^4 + \left(\frac{3}{4}\right)^8 + \dots \right]$$

$$= \frac{1/4}{1 - \left(\frac{3}{4}\right)^4} = \frac{1}{4} \times \frac{256}{175} = \frac{64}{175}.$$

B can cut a spade in the second, sixth, tenth, ..., drawings, the respective probabilities of which are $\frac{3}{4} \times \frac{1}{4}, \left(\frac{3}{4}\right)^5 \times \frac{1}{4}, \left(\frac{3}{4}\right)^9 \times \frac{1}{4}, \dots$

$$\begin{aligned}
 \text{Therefore } P(B \text{ cuts a spade first}) &= \frac{3}{4} \cdot \frac{1}{4} + \left(\frac{3}{4}\right)^5 \cdot \frac{1}{4} + \left(\frac{3}{4}\right)^9 \cdot \frac{1}{4} + \dots \\
 &= \frac{3}{4} \cdot \frac{1}{4} \left[1 + \left(\frac{3}{4}\right)^4 + \left(\frac{3}{4}\right)^8 + \dots \right] \\
 &= \frac{\frac{3}{4} \times \frac{1}{4}}{1 - \left(\frac{3}{4}\right)^4} = \frac{3}{16} \times \frac{256}{175} = \frac{48}{175}
 \end{aligned}$$

C can cut a spade in the third, seventh, eleventh, ..., drawings, the respective probabilities of which are

$$\left(\frac{3}{4}\right)^2 \cdot \frac{1}{4}, \left(\frac{3}{4}\right)^6 \cdot \frac{1}{4}, \left(\frac{3}{4}\right)^{10} \cdot \frac{1}{4}, \dots \text{Thus}$$

$$\begin{aligned}
 P(C \text{ cuts a spade first}) &= \left(\frac{3}{4}\right)^2 \cdot \frac{1}{4} + \left(\frac{3}{4}\right)^6 \cdot \frac{1}{4} + \left(\frac{3}{4}\right)^{10} \cdot \frac{1}{4} + \dots \\
 &= \left(\frac{3}{4}\right)^2 \cdot \frac{1}{4} \left[1 + \left(\frac{3}{4}\right)^4 + \left(\frac{3}{4}\right)^8 + \dots \right] \\
 &= \frac{\left(\frac{3}{4}\right)^2 \cdot \frac{1}{4}}{1 - \left(\frac{3}{4}\right)^4} = \frac{9}{64} \times \frac{256}{175} = \frac{36}{175}
 \end{aligned}$$

$$\begin{aligned}
 \text{and } P(D \text{ cuts a spade first}) &= \frac{64}{175} + \frac{48}{175} + \frac{36}{175} \\
 &= 1 - \frac{148}{175} = \frac{27}{175}
 \end{aligned}$$

$$\text{Hence expectation of A} = \frac{64}{175} \times \text{£ } 175 = \text{£ } 64:$$

$$\text{expectation of B} = \frac{48}{175} \times \text{£ } 175 = \text{£ } 48:$$

$$\text{expectation of C} = \frac{36}{175} \times \text{£ } 175 = \text{£ } 36: \text{ and}$$

$$\text{expectation of D} = \frac{27}{175} \times \text{£ } 175 = \text{£ } 27.$$

(c) The bag contains 2 white and 2 black balls.

Let A denote the event that A draws a white ball first. Then

$$P(A) = \binom{2}{1} \div \binom{4}{1} = \frac{1}{2}$$

If A fails to draw the white ball, then the bag will contain 2 white and 1 black ball, as the ball drawn by A is not replaced.

Let B denote the event that B draws a white ball after A fails to draw it. Then

$$P(B) = P(B \text{ draws white ball} / A \text{ has failed to draw white ball})$$

$$= \frac{\binom{2}{1}}{\binom{3}{1}} \times \frac{2}{4} = \frac{1}{3}$$

If both A and B have failed to draw white ball, then the bag will contain 2 white balls as the balls being drawn are not replaced.

Let C denote the event that C draws a white ball given that both A and B have failed to draw it. Then

$$P(C) = \frac{2}{2} \times \frac{2}{4} \times \frac{1}{3} = \frac{1}{6}$$

$$\text{Hence expectation of } A = \frac{1}{2} \times \text{Rs. } 18 = \text{Rs. } 9.$$

$$\text{expectation of } B = \frac{1}{3} \times \text{Rs. } 18 = \text{Rs. } 6, \text{ and}$$

$$\text{expectation of } C = \frac{1}{6} \times \text{Rs. } 18 = \text{Rs. } 3$$

7.27. (a) The total area under the curve is

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^1 x^2(1-x) dx$$

$$= \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12} (=N, \text{ say})$$

Hence this is not a density function.

$$\text{Now mean} = \frac{1}{N} \int x \cdot f(x) dx$$

$$\begin{aligned} &= 12 \int_0^1 (x^3 - x^4) dx = 12 \left[\frac{x^4}{4} - \frac{x^5}{5} \right]_0^1 \\ &= 12 \left[\frac{1}{4} - \frac{1}{5} \right] = 12 \times \frac{1}{20} = \frac{3}{5} = 0.06. \end{aligned}$$

$$E(X^2) = 12 \left[\int_0^1 x^2 f(x) dx \right]$$

$$\begin{aligned} &= 12 \int_0^1 (x^4 - x^5) dx = 12 \left[\frac{x^5}{5} - \frac{x^6}{6} \right]_0^1 \\ &= 12 \left[\frac{1}{5} - \frac{1}{6} \right] = \frac{12}{30} = \frac{2}{5} = 0.4. \end{aligned}$$

Thus σ^2 or $\text{Var}(X) = E(X^2) - [E(X)]^2 = 0.4 - (0.6)^2 = 0.04$, so that

$$\sigma = \sqrt{0.04} = 0.2.$$

(b) In order that $f(x)$ may be a density function, we should have $f(x) \geq 0$ for every x , and

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

The first condition is clearly satisfied and the second condition will be satisfied if

$$c \int_1^2 x dx = 1, \text{ i.e. if } c \left[\frac{x^2}{2} \right]_1^2 = 1,$$

$$\text{i.e. if } c \left(\frac{3}{2} \right) = 1 \text{ i.e. if } c = \frac{2}{3}.$$

$$(ii) \quad \text{Mean} = E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx = \frac{2}{3} \int_1^2 x \cdot x dx$$

$$= \frac{2}{3} \int_1^2 x^2 dx = \frac{2}{3} \left[\frac{x^3}{3} \right]_1^2$$

$$= \frac{2}{3} \left[\frac{8}{3} - \frac{1}{3} \right] = \frac{2}{3} \cdot \frac{7}{3} = \frac{14}{9},$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx = \frac{2}{3} \int_1^2 x^2 \cdot x dx = \frac{2}{3} \int_1^2 x^3 dx = \frac{2}{3} \left[\frac{x^4}{4} \right]_1^2$$

$$= \frac{2}{3} \left[\frac{16}{4} - \frac{1}{4} \right] = \frac{2}{3} \times \frac{15}{4} = \frac{5}{2}.$$

$$\text{Var}(X) = E(X^2) - \{E(X)\}^2 = \frac{5}{2} - \left(\frac{14}{9}\right)^2 = \frac{5}{2} - \frac{196}{81} = \frac{13}{162},$$

$$\text{and S.D.}(X) = \sqrt{\text{Var}(X)} = \sqrt{\frac{13}{162}}.$$

$$(c) \quad E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx \quad (\text{by definition})$$

$$= \int_{-\infty}^0 x \cdot 0 dx + \frac{3}{8} \int_0^2 x (x-2)^2 dx + \int_2^{\infty} x \cdot 0 dx$$

$$= \frac{3}{8} \int_0^2 x(x^2 - 4x + 4) dx = \frac{3}{8} \int_0^2 (x^3 - 4x^2 + 4x) dx$$

$$= \frac{3}{8} \left[\frac{x^4}{4} - \frac{4x^3}{3} + \frac{4x^2}{2} \right]_0^2$$

$$= \frac{3}{8} \left[4 - \frac{32}{3} + 8 \right] = \frac{3}{8} \left(\frac{4}{3} \right) = \frac{1}{2}, \text{ and}$$

$$\sigma = \sqrt{E(X^2) - [E(X)]^2}, \text{ where}$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$= \frac{3}{8} \int_0^2 x^2 (x-2)^2 dx = \frac{3}{8} \int_0^2 (x^4 - 4x^3 + 4x^2) dx$$

$$= \frac{3}{8} \left[\frac{x^5}{5} - \frac{4x^4}{4} + \frac{4x^3}{3} \right]_0^2 = \frac{3}{8} \left[\frac{32}{5} - 16 + \frac{32}{3} \right]$$

$$= \frac{3}{8} \left(\frac{16}{15} \right) = \frac{2}{5}$$

$$\therefore \sigma = \sqrt{\frac{2}{5} - \left(\frac{1}{2}\right)^2} = \sqrt{0.4 - 0.25}$$

$$= \sqrt{0.15} = 0.387$$

7.28. (a) The function $f(x)$ will be a density function, if

$$k \int_0^2 x^3 (1-x) dx = 1, \text{ i.e. if } k \int_0^1 (x^3 - x^4) dx = 1,$$

or $k \left[\frac{x^4}{4} - \frac{x^5}{5} \right]_0^1 = 1, \text{ or } k \left[\frac{1}{4} - \frac{1}{5} \right] = 1, \text{ or } k = 20$

$\therefore f(x) = 20x^3 (1-x)$ is a proper density function.

The mean, $E(X)$ and the $\text{Var}(X)$ are determined as below:

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

$$= 20 \int_0^1 x \cdot x^3 (1-x) dx = 20 \left[\frac{x^5}{5} - \frac{x^6}{6} \right]_0^1$$

$$= 20 \left[\frac{1}{5} - \frac{1}{6} \right] = \frac{20}{30} = \frac{2}{3},$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$= 20 \int_0^1 x^2 \cdot x^3(1-x) dx = 20 \int_0^1 (x^5 - x^6) dx$$

$$= 20 \left[\frac{x^6}{6} - \frac{x^7}{7} \right]_0^1 = 20 \left(\frac{1}{42} \right) = \frac{10}{21}, \text{ and}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{10}{21} - \left(\frac{2}{3} \right)^2 = \frac{10}{21} - \frac{4}{9} = \frac{2}{63}.$$

(b) The function $f(x)$ will be a density function, if

$$\int_0^3 Ax(9-x^2) dx = 1$$

$$\text{i.e. } A \left[\frac{9x^2}{2} - \frac{x^4}{4} \right]_0^3 = 1$$

$$\text{i.e. } A \left[\frac{81}{2} - \frac{81}{4} \right] = 1, \text{ which gives } A = \frac{4}{81}.$$

$$\therefore f(x) = \frac{4}{81} x(9-x^2), \text{ for } 0 \leq x \leq 3$$

$$= 0, \text{ elsewhere}$$

$$\text{Now Mean} = E(X) = \frac{4}{81} \int_0^3 x \cdot x(9-x^2) dx$$

$$= \frac{4}{81} \left[\frac{9x^3}{3} - \frac{x^5}{5} \right]_0^3$$

$$= \frac{4}{81} \left[81 - \frac{243}{5} \right] = \frac{4}{81} \times \frac{162}{5} = \frac{8}{5}$$

$$E(X^2) = \frac{4}{81} \int_0^3 x^2 (9x - x^3) dx$$

$$= \frac{4}{81} \left[\frac{9x^4}{4} - \frac{x^6}{6} \right]_0^3 = \frac{4}{81} \left[\frac{729}{4} - \frac{729}{6} \right]$$

$$= \frac{4}{81} \times \frac{729}{12} = 3.$$

$$\therefore S.D. = \sqrt{\sum(X^2) - [E(X)]^2} = \sqrt{3 - \left(\frac{8}{5}\right)^2}$$

$$= \sqrt{3 - 2.56} = 0.66$$

7.29. The total area must be unity.

$$\frac{1}{16} \int_{-3}^{-1} (3+x)^2 dx + \frac{1}{16} \int_{-1}^{+1} (6-2x^2) dx + \frac{1}{16} \int_1^3 (3-x)^2 dx$$

$$= \frac{1}{16} \left[\frac{(3+x)^3}{3} \right]_{-3}^{-1} + \frac{1}{16} \left[6x - \frac{2x^3}{3} \right]_{-1}^{+1} + \frac{1}{16} \left[\frac{-(3-x)^3}{3} \right]_1^3$$

$$= \frac{1}{16} \left[\frac{8}{3} + \frac{32}{3} + \frac{8}{3} \right] = 1.$$

$$\text{Now } \mu = \int_{-\infty}^{\infty} xf(x) dx$$

$$= \frac{1}{16} \int_{-3}^{-1} x(3+x)^2 dx + \frac{1}{16} \int_{-1}^{+1} x(6-2x^2) dx + \frac{1}{16} \int_1^3 x(3-x)^2 dx$$

$$= \frac{1}{16} \left[\int_{-3}^{-1} (9x+6x^2+x^3) dx + \int_{-1}^1 (6x-2x^3) dx + \int_1^3 (9x-6x^2+x^3) dx \right]$$

$$= \frac{1}{16} \left[\frac{9x^2}{2} + \frac{6x^3}{3} + \frac{x^4}{4} \right]_{-3}^{-1} + \frac{1}{16} \left[\frac{6x^2}{2} - \frac{2x^4}{4} \right]_{-1}^{+1} +$$

$$\frac{1}{16} \left[\frac{9x^2}{2} - \frac{6x^3}{3} + \frac{x^4}{4} \right]_1^3$$

$$= \frac{1}{16} \left[\left(\frac{9}{2} - 2 + \frac{1}{4} \right) - \left(\frac{81}{2} - 54 + \frac{81}{4} \right) \right] +$$

$$\frac{1}{16} \left[\left(3 - \frac{1}{2} \right) - \left(3 - \frac{1}{2} \right) \right] + \frac{1}{16} \left[\left(\frac{81}{2} - 54 + \frac{81}{4} \right) - \left(\frac{9}{2} - 2 + \frac{1}{4} \right) \right]$$

$$= \frac{1}{16} \left[\frac{11}{4} - \frac{189}{4} \right] + 0 + \frac{1}{16} \left[\frac{189}{4} - \frac{11}{4} \right] = 0$$

$$\sigma^2 = \int_{-\infty}^{\infty} x^2 f(x) dx - (\mu)^2$$

$$= \frac{1}{16} \int_{-3}^{-1} x^2 (3+x)^2 dx + \frac{1}{16} \int_{-1}^{+1} x^2 (6-2x^2) dx +$$

$$\frac{1}{16} \int_1^3 x^2 (3-x)^2 dx \quad (\because \mu = 0)$$

$$= \frac{1}{16} \int_{-3}^{-1} (9x^2 + 6x^3 + x^4) dx + \frac{1}{16} \int_{-1}^{+1} (6x^2 - 2x^4) dx +$$

$$\frac{1}{16} \int_1^3 (9x^2 - 6x^3 + x^4) dx$$

$$= \frac{1}{16} \left[3x^3 + \frac{3x^4}{2} + \frac{x^5}{5} \right]_{-3}^{-1} + \frac{1}{16} \left[2x^3 - \frac{2x^5}{5} \right]_{-1}^{+1} +$$

$$\frac{1}{16} \left[3x^3 - \frac{3x^4}{2} + \frac{x^5}{5} \right]_1^3$$

$$= \frac{1}{16} \left[\left(-3 + \frac{3}{2} - \frac{1}{5} \right) - \left(-81 + \frac{243}{2} - \frac{243}{5} \right) \right] +$$

$$\frac{1}{16} \left[\left(2 - \frac{2}{5} \right) - \left(-2 + \frac{2}{5} \right) \right] +$$

$$\frac{1}{16} \left[\left(81 - \frac{243}{2} + \frac{243}{5} \right) - \left(3 - \frac{3}{2} + \frac{1}{5} \right) \right]$$

$$= \frac{1}{16} \left[\frac{81}{10} - \frac{17}{10} \right] + \frac{1}{16} \left[\frac{8}{5} + \frac{8}{5} \right] + \frac{1}{16} \left[\frac{81}{10} - \frac{17}{10} \right]$$

$$= \frac{2}{5} + \frac{1}{5} + \frac{2}{5} = 1$$

Hence $\sigma = \sqrt{1} = 1$

7.30. The function $f(x)$ will be a proper density function, if

$$k \int_0^3 (6 + x - x^2) dx = 1,$$

$$\text{i.e. if } k \left[6x + \frac{x^2}{2} - \frac{x^3}{3} \right]_0^3 = 1, \text{ i.e. if } k \left[18 + \frac{9}{2} - \frac{27}{3} \right] = 1,$$

$$\text{i.e. if } k \left[\frac{27}{2} \right] = 1, \text{ i.e. if } k = \frac{2}{27}.$$

Now $\mu = E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx$

$$= \frac{2}{27} \int_0^3 x \cdot (6 + x - x^2) dx = \frac{2}{27} \int_0^3 (6x + x^2 - x^3) dx$$

$$= \frac{2}{27} \left[\frac{6x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} \right]_0^3 = \frac{2}{27} \left[27 + 9 - \frac{81}{4} \right]$$

$$= \frac{2}{27} \left(\frac{63}{4} \right) = \frac{7}{6}.$$

For the mode of the distribution, $f'(x) = 0$ and $f''(x) < 0$.

$$\text{Now } f(x) = \frac{2}{27} (6 + x - x^2)$$

$$\therefore f'(x) = \frac{2}{27} (1 - 2x), \text{ and } f''(x) = \frac{-4}{27} < 0.$$

$$\text{Thus } f'(x) = 0 \text{ gives } x = \frac{1}{2}$$

Hence Mode = $\frac{1}{2}$.

Again $\text{Var}(X) = E(X^2) - [E(X)]^2$, where

$$\begin{aligned} E(X^2) &= \frac{2}{27} \int_0^3 x^2(6 + x - x^2) dx = \frac{2}{27} \int_0^3 (6x^2 + x^3 - x^4) dx \\ &= \frac{2}{27} \left[\frac{6x^3}{3} + \frac{x^4}{4} - \frac{x^5}{5} \right]_0^3 = \frac{2}{27} \left[54 + \frac{81}{4} - \frac{243}{5} \right] \\ &= \frac{2}{27} \left[\frac{513}{20} \right] = \frac{19}{10}. \text{ Therefore} \end{aligned}$$

$$\sigma^2 = \frac{19}{10} - \left(\frac{7}{6} \right)^2 = \frac{19}{10} - \frac{49}{36} = \frac{97}{180}.$$

7.31. Since $f(x)$ is a probability density function, therefore

$$k \int_2^5 (2-x)(x-5) dx = 1.$$

We find the value of k as follows:

$$k \int_2^5 (2-x)(x-5) dx = 1$$

or $k \int_2^5 (-x^2 + 7x - 10) dx = 1$

i.e. $k \left[-\frac{x^3}{3} + \frac{7x^2}{2} - 10x \right]_2^5 = 1$

i.e. $k \left[\left(-\frac{125}{3} + \frac{175}{2} - 50 \right) - \left(-\frac{8}{3} + 14 - 20 \right) \right] = 1$

i.e. $k \left[-\frac{25}{6} + \frac{26}{3} \right] = 1$, i.e. $k \left(\frac{9}{2} \right) = 1$ giving $k = 2/9$

Now the mean of X is

$$\begin{aligned} E(X) &= \frac{2}{9} \int_2^5 x(2-x)(x-5) dx \\ &= \frac{2}{9} \int_2^5 (-x^3 + 7x^2 - 10x) dx \\ &= \frac{2}{9} \left[-\frac{x^4}{4} + \frac{7x^3}{3} - \frac{10x^2}{2} \right]_2^5 \\ &= \frac{2}{9} \left[\left(-\frac{625}{4} + \frac{875}{3} - \frac{250}{2} \right) - \left(-4 + \frac{56}{3} - 20 \right) \right] \\ &= \frac{2}{9} \left[\frac{125}{12} + \frac{16}{3} \right] = \frac{2}{9} \left(\frac{189}{12} \right) = \frac{7}{2}, \text{ and} \end{aligned}$$

$\text{Var}(X) = E(X^2) - [E(X)]^2$, where

$$\begin{aligned} E(X^2) &= \int_2^5 x^2 f(x) dx = \frac{2}{9} \int_2^5 x^2 (2-x)(x-5) dx \\ &= \frac{2}{9} \int_2^5 (-x^4 + 7x^3 - 10x^2) dx \\ &= \frac{2}{9} \left[-\frac{x^5}{5} + \frac{7x^4}{4} - \frac{10x^3}{3} \right]_2^5 = \frac{2}{9} \left(\frac{1143}{20} \right) = \frac{127}{10} \end{aligned}$$

$$\therefore \text{Var}(X) = \frac{127}{10} - \left(\frac{7}{2} \right)^2 = \frac{9}{20}$$

For the mode of the distribution, $f'(x) = 0$ and $f''(x) < 0$.

$$\text{Now } f(x) = \frac{2}{9} (-x^2 + 7x - 10)$$

$$\therefore f'(x) = \frac{2}{9} (-2x + 7), \text{ and}$$

$$f''(x) = \frac{2}{9}(-2) = \frac{-4}{9}, \text{ which is negative.}$$

Thus $f'(x) = 0$ gives $x = \frac{7}{2}$

Hence mode of the distribution is at $x = \frac{7}{2}$.

The median, a , is given by $\int_{-\infty}^a f(x) dx = \frac{1}{2}$. Thus

$$\frac{2}{9} \int_2^a (2-x)(x-5) dx = \frac{1}{2}$$

or $\frac{2}{9} \left[-\frac{x^3}{3} + \frac{7x^2}{2} - 10x \right]_2^a = \frac{1}{2}$

or $\frac{2}{9} \left[-\frac{a^3}{3} + \frac{7a^2}{2} - 10a + \frac{26}{3} \right] = \frac{1}{2}$

This equation reduces to

$$4a^3 - 42a^2 + 120a - 77 = 0.$$

Since the probability density function, which is parabolic, is symmetrical, so the equation has solution $a = 7/2$ and thus we may factorize the equation given above as

$$(2a - 7)(2a^2 - 14a + 11) = 0$$

Now $2a - 7 = 0$ gives $a = \frac{7}{2}$ and

$$2a^2 - 14a + 11 = 0 \text{ gives } a = 0.902 \text{ and } 6.098$$

But both the values 0.902 and 6.098 are unacceptable since a must lie in the interval (2, 5). The median is therefore given by $a = 7/2$.

7.32. By definition, we have

$$\log G = E [\log X] = \int_{-\infty}^{\infty} \log x f(x) dx, \text{ if it exists.}$$

$$= 6 \int_1^2 (\log x) (2-x) (x-1) dx$$

$$= 6 \int_1^2 (\log x) (-x^2 + 3x - 2) dx$$

$$= -6 \int_1^2 x^2 \log x dx + 18 \int_1^2 x \log x dx$$

$$-12 \int_1^2 \log x dx$$

To evaluate $\int_1^2 x^2 \log x dx$, we put $u = \log x$ and $dv = x^2 dx$ so

that $du = \frac{1}{x}$ and $v = \frac{x^3}{3}$. Thus integrating by parts, we have

$$\int_1^2 x^2 \log x dx = \left[\frac{x^3}{3} \log x \right]_1^2 - \int_1^2 \frac{x^3}{3} \cdot \frac{1}{x} dx$$

$$\left[\because \int_0^t u dv = uv \Big|_0^t - \int_0^t v du \right]$$

$$= \left[\frac{8}{3} \log 2 - 0 \right] - \frac{1}{3} \left[\frac{x^3}{3} \right]_1^2 = \frac{8}{3} \log 2 - \frac{7}{9};$$

$$\int_1^2 x \log x dx = \left[\frac{x^2}{2} \log x \right]_1^2 - \int_1^2 \frac{x^2}{2} \cdot \frac{1}{x} dx$$

(Integrating by parts)

$$= 2 \log 2 - \frac{1}{2} \left[\frac{x^2}{2} \right]_1^2 = 2 \log 2 - \frac{3}{4}$$

$$\int_1^2 x \log x \, dx = \left[x \log x \right]_1^2 - \int_1^2 x \cdot \frac{1}{x} dx$$

(Integrating by parts)

$$= 2 \log 2 - \left[x \right]_1^2 = 2 \log 2 - 1.$$

$$\text{Thus } \log G = -6 \left[\frac{8}{3} \log 2 - \frac{7}{9} \right] + 18 \left[2 \log 2 - \frac{3}{4} \right] - 12 [2 \log 2 - 1]$$

$$= -4 \log 2 + \frac{19}{6} = -\log 2^4 + \frac{19}{6}$$

$$\text{or } \log G + \log 16 = \frac{19}{6} \text{ or } \log (16G) = \frac{19}{6}$$

$$\text{or } 6 \log (16G) = 19.$$

7.33. A function $f(x)$ is a proper density function, if

$$\int_{-\infty}^{\infty} f(x) \, dx = 1. \text{ Therefore}$$

$$k \int_0^1 x^2(1-x)^3 \, dx = 1$$

Using the given relationship $\int_0^1 x^m(1-x)^n \, dx = \frac{m!n!}{(m+n+1)!}$, we get

$$k \left[\frac{2! 3!}{(2+3+1)!} \right] = 1, \text{ or } k = 60.$$

Hence $f(x) = 60x^2(1-x)^3$, where $0 \leq x \leq 1$, is a proper *p.d.f.*

The skewness, in this case, is measured by μ_3 , which in symmetrical distribution is zero.

$$\text{Now } \mu'_1 = E(X) = \int_{-\infty}^{\infty} x \cdot f(x) \, dx$$

$$= 60 \int_0^1 x \cdot x^2(1-x)^3 \, dx = 60 \int_0^1 x^3(1-x)^3 \, dx$$

Using the relationship $\int_0^1 x^m(1-x)^n dx = \frac{m! n!}{(m+n+1)!}$, we get

$$\mu_1' = 60 \cdot \frac{3! 3!}{(3+3+1)!} = 60 \left(\frac{1}{140} \right) = \frac{3}{7}.$$

$$\begin{aligned} \mu_2' = E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx = 60 \int_0^1 x^2 \cdot x^2(1-x)^3 dx \\ &= 60 \int_0^1 x^4(1-x)^3 dx = 60 \cdot \frac{4! 3!}{(4+3+1)!} = \frac{3}{14}, \text{ and} \end{aligned}$$

$$\begin{aligned} \mu_3' = E(X^3) &= \int_{-\infty}^{\infty} x^3 f(x) dx = 60 \int_0^1 x^5(1-x)^3 dx \\ &= 60 \frac{5! 3!}{(5+3+1)!} = \frac{5}{42}. \end{aligned}$$

$$\begin{aligned} \text{Hence } \mu_3 &= \mu_3' - 3\mu_1' \mu_2' + 2\mu_1'^3 \\ &= \frac{5}{42} - 3 \left(\frac{3}{7} \right) \left(\frac{3}{14} \right) + 2 \left(\frac{3}{7} \right)^3 \\ &= \frac{5}{42} - \frac{27}{98} + \frac{54}{343} = \frac{2}{2058} = 0.001. \end{aligned}$$

7.34 (a) First of all, we find the value of k , which should be such as to make

$$\int_{-a}^a k dx = 1$$

$$\text{or } k \left[x \right]_{-a}^a = 1 \text{ or } 2ak = 1 \text{ or } k = \frac{1}{2a}$$

Now, the mean, μ , is given by

$$\mu = \int_{-a}^a x \cdot \frac{1}{2a} dx = \frac{1}{2a} \left[\frac{x^2}{2} \right]_{-a}^a = \frac{1}{2a} (0) = 0.$$

The variance, σ^2 or μ_2 , is given by

$$\begin{aligned}\sigma^2 &= \frac{1}{2a} \int_{-a}^a x^2 dx - (\mu)^2 = \frac{1}{2a} \left[\frac{x^3}{3} \right]_{-a}^a \quad (\because \mu = 0) \\ &= \frac{1}{2a} \left[\frac{a^3}{3} + \frac{a^3}{3} \right] = \frac{a^2}{3}.\end{aligned}$$

Again Mean Deviation about the mean is

$$\begin{aligned}M.D. &= \int_{-\infty}^{\infty} |x - \mu| f(x) dx = \frac{1}{2a} \int_{-a}^a |x - \mu| dx \\ &= \frac{1}{2a} \left\{ \int_{-a}^0 -x dx + \int_0^a x dx \right\} \quad (\because \mu = 0) \\ &= \frac{2}{2a} \int_0^a x dx = \frac{1}{a} \left[\frac{x^2}{2} \right]_0^a = \frac{a}{2}.\end{aligned}$$

(b) Calculation of the mean moments and the mean deviation.

$$\mu = \int_{-\infty}^{\infty} x f(x) dx = \int_0^1 x dx = \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2}.$$

Now, $\mu_1 = 0$;

$$\begin{aligned}\mu_2 &= \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \\ &= \int_0^1 \left(x - \frac{1}{2}\right)^2 dx = \int_0^1 \left(x^2 - x + \frac{1}{4}\right) dx \\ &= \left[\frac{x^3}{3} - \frac{x^2}{2} + \frac{x}{4} \right]_0^1 = \frac{1}{3} - \frac{1}{2} + \frac{1}{4} = \frac{1}{12};\end{aligned}$$

$$\begin{aligned}
 \mu_3 &= \int_{-\infty}^{\infty} (x - \mu)^3 f(x) dx \\
 &= \int_0^1 \left(x - \frac{1}{2}\right)^3 dx = \int_0^1 \left(x^3 - \frac{3x^2}{2} + \frac{3x}{4} - \frac{1}{8}\right) dx \\
 &= \left[\frac{x^4}{4} - \frac{x^3}{2} + \frac{3x^2}{8} - \frac{x}{8} \right]_0^1 = \frac{1}{4} - \frac{1}{2} + \frac{3}{8} - \frac{1}{8} = \frac{0}{8} = 0;
 \end{aligned}$$

$$\begin{aligned}
 \mu_4 &= \int_{-\infty}^{\infty} (x - \mu)^4 f(x) dx = \int_0^1 \left(x - \frac{1}{2}\right)^4 dx \\
 &= \int_0^1 \left(x^4 - 2x^3 + \frac{3x^2}{2} - \frac{x}{2} + \frac{1}{16}\right) dx \\
 &= \left[\frac{x^5}{5} - \frac{x^4}{2} + \frac{x^3}{2} - \frac{x^2}{4} + \frac{x}{16} \right]_0^1 \\
 &= \frac{1}{5} - \frac{1}{2} + \frac{1}{2} - \frac{1}{4} + \frac{1}{16} = \frac{1}{80}.
 \end{aligned}$$

The Mean Deviation about the mean is

$$\begin{aligned}
 M.D. &= \int_{-\infty}^{\infty} |x - \mu| f(x) dx = \int_0^1 |x - 1/2| dx \\
 &= \int_0^{1/2} \left(\frac{1}{2} - x\right) dx + \int_{1/2}^1 \left(x - \frac{1}{2}\right) dx \\
 &= \left[\frac{x}{2} - \frac{x^2}{2} \right]_0^{1/2} + \left[\frac{x^2}{2} - \frac{x}{2} \right]_{1/2}^1 \\
 &= \left(\frac{1}{4} - \frac{1}{8}\right) + \left\{ \left(\frac{1}{2} - \frac{1}{2}\right) - \left\{ \left(\frac{1}{8} - \frac{1}{4}\right) \right\} \right\} \\
 &= \frac{1}{8} + \frac{1}{8} = \frac{1}{4}.
 \end{aligned}$$

7.35. The total probability must be unity.

$$\begin{aligned}
 \int_{-\infty}^{\infty} f(x) dx &= \frac{1}{2} \int_0^1 x dx + \frac{1}{2} \int_1^2 dx + \frac{1}{2} \int_2^3 (3-x) dx \\
 &= \frac{1}{2} \left[\frac{x^2}{2} \right]_0^1 + \frac{1}{2} \left[x \right]_1^2 + \frac{1}{2} \left[\frac{-(3-x)^2}{2} \right]_2^3 \\
 &= \frac{1}{2} \left(\frac{1}{2} \right) + \frac{1}{2} (1) + \frac{1}{2} \left(\frac{1}{2} \right) = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1.
 \end{aligned}$$

Now $\mu = \int_{-\infty}^{\infty} x f(x) dx$

$$\begin{aligned}
 &= \frac{1}{2} \int_0^1 x^2 dx + \frac{1}{2} \int_1^2 x^2 dx + \frac{1}{2} \int_2^3 x(3-x) dx \\
 &= \frac{1}{2} \left[\frac{x^3}{3} \right]_0^1 + \frac{1}{2} \left[\frac{x^3}{3} \right]_1^2 + \frac{1}{2} \left[\frac{3x^2}{2} - \frac{x^3}{3} \right]_2^3 \\
 &= \frac{1}{2} \left(\frac{1}{3} \right) + \frac{1}{2} \left[2 - \frac{1}{3} \right] + \frac{1}{2} \left[\left(\frac{27}{2} - \frac{27}{3} \right) - \left(6 - \frac{8}{3} \right) \right] \\
 &= \frac{1}{6} + \frac{3}{4} + \frac{7}{12} = \frac{18}{12} = \frac{3}{2}.
 \end{aligned}$$

$\sigma^2 = \mu'_2 - (\mu)^2 = \int_{-\infty}^{\infty} x^2 f(x) dx - (\mu)^2$

$$\begin{aligned}
 &= \left\{ \frac{1}{2} \int_0^1 x^3 dx + \frac{1}{2} \int_1^2 x^2 dx + \frac{1}{2} \int_2^3 x^2(3-x) dx \right\} - \left(\frac{3}{2} \right)^2 \\
 &= \left\{ \frac{1}{2} \left[\frac{x^4}{4} \right]_0^1 + \frac{1}{2} \left[\frac{x^3}{3} \right]_1^2 + \frac{1}{2} \left[x^3 - \frac{x^4}{4} \right]_2^3 \right\} - \frac{9}{4} \\
 &= \left\{ \frac{1}{2} \left(\frac{1}{4} \right) + \frac{1}{2} \left(\frac{8}{3} - \frac{1}{3} \right) + \frac{1}{2} \left(\frac{27}{4} - \frac{4}{1} \right) \right\} - \frac{9}{4} \\
 &= \left\{ \frac{1}{8} + \frac{7}{6} + \frac{11}{8} \right\} - \frac{9}{4} = \frac{8}{3} - \frac{9}{4} = \frac{5}{12}.
 \end{aligned}$$

In order to find β_2 , the moment measure of Kurtosis, we first find the third and fourth moments. Thus

$$\begin{aligned}\mu'_3 &= \int_{-\infty}^{\infty} x^3 \cdot f(x) dx \\&= \frac{1}{2} \int_0^1 x^4 dx + \frac{1}{2} \int_1^2 x^3 dx + \frac{1}{2} \int_2^3 x^3(3-x) dx \\&= \frac{1}{2} \left[\frac{x^5}{5} \right]_0^1 + \frac{1}{2} \left[\frac{x^4}{4} \right]_1^2 + \frac{1}{2} \left[\frac{3x^4}{4} - \frac{x^5}{5} \right]_2^3 \\&= \frac{1}{10} + \frac{15}{8} + \frac{1}{2} \left(\frac{243}{20} - \frac{28}{5} \right) = \frac{1}{10} + \frac{15}{8} + \frac{131}{40} = \frac{21}{4}\end{aligned}$$

$$\begin{aligned}\mu'_4 &= \int_{-\infty}^{\infty} x^4 \cdot f(x) dx \\&= \frac{1}{2} \int_0^1 x^5 dx + \frac{1}{2} \int_1^2 x^4 dx + \frac{1}{2} \int_2^3 x^4(3-x) dx \\&= \frac{1}{2} \left[\frac{x^6}{6} \right]_0^1 + \frac{1}{2} \left[\frac{x^5}{5} \right]_1^2 + \frac{1}{2} \left[\frac{3x^5}{5} - \frac{x^6}{6} \right]_2^3 \\&= \frac{1}{2} \left(\frac{1}{6} \right) + \frac{1}{2} \left(\frac{31}{5} \right) + \frac{1}{2} \left(\frac{729}{30} - \frac{128}{15} \right) \\&= \frac{1}{12} + \frac{31}{10} + \frac{473}{60} = \frac{166}{15}\end{aligned}$$

Now $\mu_3 = \mu'_3 - 3\mu'_1 \mu'_2 + 2(\mu'_1)^3$

$$= \frac{21}{4} - 3 \left(\frac{3}{2} \right) \left(\frac{8}{3} \right) + 2 \left(\frac{3}{2} \right)^3 = \frac{21}{4} - \frac{12}{1} + \frac{27}{4} = 0, \text{ and}$$

$$\begin{aligned}\mu_4 &= \mu'_4 - 4\mu'_1 \mu'_3 + 6\mu'^2_1 \mu'_2 - 3\mu'^4_1 \\&= \frac{166}{15} - 4 \left(\frac{3}{2} \right) \left(\frac{21}{4} \right) + 6 \left(\frac{3}{2} \right)^2 \left(\frac{8}{3} \right) - 3 \left(\frac{3}{2} \right)^4\end{aligned}$$

$$= \frac{166}{15} - \frac{63}{2} + \frac{36}{1} - \frac{243}{16} = \frac{91}{240}$$

Hence $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{91}{240} \times \frac{144}{25} = \frac{273}{125} = 2.184.$

7.36. Here the distribution is $f(x) = x^2(6-x)^2$ between $x=0$ and $x=6$. Therefore we have

$$\begin{aligned} \int_0^6 x^2(6-x)^2 dx &= \int_0^6 (x^4 - 12x^3 + 36x^2) dx \\ &= \left[\frac{x^5}{5} - \frac{12x^4}{4} + \frac{36x^3}{3} \right]_0^6 \\ &= \frac{(6)^5}{5} - 3(6)^4 + 12(6)^3 \\ &= 216 \left[\frac{36}{5} - 18 + 12 \right] = 216 \times \frac{6}{5} = \frac{1296}{5} \end{aligned}$$

To make the total frequency unity, we must multiply $f(x)$ by $\frac{5}{1296}$.

Now we calculate the moments about origin as:

$$\begin{aligned} \mu'_1 &= \int_{-\infty}^{\infty} xf(x) dx = \frac{5}{1296} \int_0^6 x^3(x^2 - 12x + 36) dx \\ &= \frac{5}{1296} \left[\frac{x^6}{6} - \frac{12x^5}{5} + \frac{36x^4}{4} \right]_0^6 \\ &= \frac{5}{1296} \left[(6)^4 \left\{ \frac{36}{6} - \frac{72}{5} + 9 \right\} \right] = \frac{5 \times 1296}{1296} \left\{ \frac{3}{5} \right\} = 3; \\ \mu'_2 &= \int_{-\infty}^{\infty} x^2 f(x) dx = \frac{5}{1296} \int_0^6 x^4(x^2 - 12x + 36) dx \end{aligned}$$

$$= \frac{5}{1296} \left[\frac{x^7}{7} - 2x^6 + \frac{36x^5}{5} \right]_0^6$$

$$= \frac{5}{1296} \times (6)^5 \left[\frac{36}{7} - 12 + \frac{36}{5} \right] = 30 \left[\frac{12}{35} \right] = \frac{72}{7};$$

$$\mu'_3 = \int_{-\infty}^{\infty} x^3 f(x) dx = \frac{5}{1296} \int_0^6 x^5(x^2 - 12x + 36) dx$$

$$= \frac{5}{1296} \left[\frac{x^8}{8} - \frac{12x^7}{7} + 6x^6 \right]_0^6$$

$$= \frac{5}{1296} \left[x^6 \left(\frac{x^2}{8} - \frac{12x}{7} + 6 \right) \right]_0^6$$

$$= \frac{5}{1296} \left[1296 \times 36 \left(\frac{9}{2} - \frac{72}{7} + 6 \right) \right] = 180 \times \frac{3}{14} = \frac{270}{7}$$

$$\mu'_4 = \int_{-\infty}^{\infty} x^4 f(x) dx = \frac{5}{1296} \int_0^6 x^6(x^2 - 12x + 36) dx$$

$$= \frac{5}{1296} \left[\frac{x^9}{9} - \frac{12x^8}{8} + \frac{36x^7}{7} \right]_0^6$$

$$= \frac{5}{1296} \left[x^7 \left(\frac{x^2}{9} - \frac{3x}{2} + \frac{36}{7} \right) \right]_0^6$$

$$= \frac{5}{1296} \left[(1296 \times 216) \left(4 - 9 + \frac{36}{7} \right) \right]$$

$$= 1080 \times \frac{1}{7} = \frac{1080}{7}.$$

Hence the moments about the mean are:

$$\mu_1 = 0;$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2 = \frac{72}{7} - (3)^2 = \frac{9}{7} = 1.29;$$

$$\begin{aligned}\mu_3 &= \mu'_3 - 3\mu'_1 \mu'_2 + 2(\mu'_1)^3 \\ &= \frac{270}{7} - 3(3) \left(\frac{72}{7} \right) + 2(3)^3 = \frac{270}{7} - \frac{648}{7} + 54 = 0;\end{aligned}$$

$$\begin{aligned}\mu_4 &= \mu'_4 - 4\mu'_1 \mu'_3 + 6(\mu'_1)^2 \mu'_2 - 3(\mu'_1)^4 \\ &= \frac{1080}{7} - \frac{3240}{7} + \frac{3888}{7} - 243 = \frac{27}{7} = 3.86;\end{aligned}$$

and the kurtosis of the distribution is

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{27}{7} \times \left(\frac{7}{9} \right)^2 = \frac{7}{3} = 2.33.$$

7.37. To compute the marginal p.d.'s and the correlation co-efficient, the values are arranged in the tabular form as below:

$X \backslash Y$	1	2	3	$g(x)$
1	$\frac{2}{15}$	$\frac{4}{15}$	$\frac{3}{15}$	$\frac{9}{15}$
2	$\frac{1}{15}$	$\frac{1}{15}$	$\frac{4}{15}$	$\frac{6}{15}$
$h(y)$	$\frac{3}{15}$	$\frac{5}{15}$	$\frac{7}{15}$	1

The correlation co-efficient, ρ , is computed as

$$\rho = \frac{E(XY) - E(X) E(Y)}{\sqrt{[E(X^2) - \{E(X)\}^2] [E(Y^2) - \{E(Y)\}^2]}} \text{, where}$$

$$E(X) = \sum_{i=1}^2 x_i g(x) = 1 \times \frac{9}{15} + 2 \times \frac{6}{15} = \frac{9}{15} + \frac{12}{15} = \frac{21}{15};$$

$$E(Y) = \sum_{j=1}^3 y_j h(y_j) = 1 \times \frac{3}{15} + 2 \times \frac{5}{15} + 3 \times \frac{7}{15}$$

$$= \frac{3}{15} + \frac{10}{15} + \frac{21}{15} = \frac{34}{15};$$

$$E(X^2) = \sum_{i=1}^2 x_i^2 g(x_i) = 1 \times \frac{9}{15} + 4 \times \frac{6}{15} = \frac{9}{15} + \frac{24}{15} = \frac{33}{15};$$

$$E(Y^2) = \sum_{j=1}^3 y_j^2 h(y_j) = 1 \times \frac{3}{15} + 4 \times \frac{5}{15} + 9 \times \frac{7}{15}$$

$$= \frac{3}{15} + \frac{20}{15} + \frac{63}{15} = \frac{86}{15}; \text{ and}$$

$$E(XY) = \sum_i \sum_j x_i y_j f(x_i, y_j) = 1 \times \frac{2}{15} + 2 \times \frac{4}{15} + 3 \times \frac{3}{15}$$

$$+ 2 \times \frac{1}{15} + 4 \times \frac{1}{15} + 6 \times \frac{1}{15}$$

$$= \frac{2}{15} + \frac{8}{15} + \frac{9}{15} + \frac{2}{15} + \frac{4}{15} + \frac{24}{15} = \frac{49}{15}$$

$$\therefore \rho = \frac{\frac{49}{15} - \left(\frac{21}{15}\right)\left(\frac{34}{15}\right)}{\sqrt{\left\{\frac{33}{15} - \left(\frac{21}{15}\right)^2\right\} \left\{\frac{86}{15} - \left(\frac{34}{15}\right)^2\right\}}} = \frac{\frac{49}{15} - \left(\frac{714}{225}\right)}{\sqrt{\left\{\frac{33}{15} - \frac{441}{225}\right\} \left\{\frac{86}{15} - \frac{1156}{225}\right\}}}$$

$$= \frac{\frac{735 - 714}{225}}{\sqrt{\left\{\frac{495 - 441}{225}\right\} \left\{\frac{1290 - 1156}{225}\right\}}} = \frac{\frac{21}{225}}{\sqrt{\left(\frac{54}{225}\right) \left(\frac{134}{225}\right)}}$$

$$= \frac{21}{85.06} = 0.25$$

7.38. (b) Given $f(x, y) = 2 - x - y$, $0 \leq x \leq 1$, $0 \leq y \leq 1$
 $= 0$, elsewhere.

Now $g(x) = \int_0^1 (2 - x - y) dy = \left[2y - xy - \frac{y^2}{2} \right]_0^1 = \frac{3}{2} - x$; and

$$h(x) = \int_0^1 (2-x-y) dx = \left[2x - \frac{x^2}{2} - xy \right]_0^1 = \frac{3}{2} - y;$$

$$E(X) = \int_0^1 x \left(\frac{3}{2} - x \right) dx = \left[\frac{3x^2}{4} - \frac{x^3}{3} \right]_0^1 = \frac{5}{12};$$

$$E(Y) = \int_0^1 y \left(\frac{3}{2} - y \right) dy = \left[\frac{3y^2}{4} - \frac{y^3}{3} \right]_0^1 = \frac{5}{12};$$

$$E(X^2) = \int_0^1 x^2 \left(\frac{3}{2} - x \right) dx = \left[\frac{3x^3}{6} - \frac{x^4}{4} \right]_0^1 = \frac{1}{4};$$

$$E(Y^2) = \int_0^1 y^2 \left(\frac{3}{2} - y \right) dy = \left[\frac{3y^3}{6} - \frac{y^4}{4} \right]_0^1 = \frac{1}{4};$$

$$\begin{aligned} E(XY) &= \int_0^1 \int_0^1 xy(2-x-y) dx dy = \int_0^1 \left[\frac{2xy^2}{2} - \frac{x^2y^2}{2} - \frac{xy^3}{3} \right]_0^1 dx \\ &= \int_0^1 \left(x - \frac{x^2}{2} - \frac{xy^3}{3} \right) dx = \left[\frac{x^2}{2} - \frac{x^3}{6} - \frac{xy^3}{3} \right]_0^1 = \frac{1}{6}. \end{aligned}$$

$$\begin{aligned} \therefore \rho &= \frac{E(XY) - E(X)E(Y)}{\sqrt{[E(X^2) - \{E(X)\}^2][E(Y^2) - \{E(Y)\}^2]}} \\ &= \frac{\frac{1}{6} - \left(\frac{5}{12}\right)\left(\frac{5}{12}\right)}{\sqrt{\left[\frac{1}{4} - \left(\frac{5}{12}\right)^2\right]\left[\frac{1}{4} - \left(\frac{5}{12}\right)^2\right]}} \\ &= \frac{\frac{1}{6} - \frac{25}{144}}{\frac{1}{4} - \frac{25}{144}} = \frac{-\frac{1}{144}}{\frac{11}{144}} = -\frac{1}{11} = -0.091. \end{aligned}$$

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